ASIA | South-East☆

I Metcalfe, University of New England, Armidale, NSW, Australia

© 2013 Elsevier Inc. All rights reserved.

Introduction	1
Origins of the South-East Asian Terranes	3
Rifting and Separation of South-East Asian Terranes from Gondwana	6
Devonian Rifting and Separation	6
Carboniferous–Permian Rifting and Early Permian Separation	6
Late Triassic to Late Jurassic Rifting and Separation	7
Amalgamation and Accretion of Terranes	7
Geological and Tectonic Evolution of South-East Asia	8
Proterozoic (2500–541 Ma)	8
Phanerozoic (541–0 Ma)	9
Cambrian–Ordovician–Silurian (541–419 Ma)	9
Devonian (419–359 Ma)	9
Carboniferous (359–299 Ma)	10
Early Permian (299–272 Ma)	10
Late Permian (272–252 Ma)	14
Triassic (252–201 Ma)	14
Jurassic (201–145 Ma)	15
Cretaceous (145–66 Ma)	16
Cenozoic (66–0 Ma)	16
South-East Asian Geological Resources	17
Oil and Gas	17
Oil- and gas-bearing basins	18
Minerals	19
Mineralization associated with ophiolites	19
Mineralization associated with volcanic arcs	19
Non-volcanic epithermal deposits	19
Tungsten deposits	20
Tin deposits	20
Mississippi valley-type epigenetic deposits	27
Iron ore deposits	27
Coal and lignite deposits	27
Diamonds, sapphires, and other gems	28
References	28

Introduction

East and South-east Asia is a giant 'jigsaw puzzle' of allochthonous continental lithospheric blocks and fragments (terranes) that are bounded by suture zones (representing the remnants of closed ocean basins) or by geological discontinuities such as major strike-slip faults (Metcalfe, 1988, 2011, 2013; and references therein) (**Figure 1**).

The complex assemblage of South-east Asian continental terranes, accretionary complexes, ophiolites, volcanic arcs, and marginal ocean basins (Hutchison, 1989) occurs in the zone of convergence between the Eurasian, Indo-Australian, and Pacific plates (**Figures 2** and **3**). In this region, two important biogeographical boundaries are recognized: the extant Wallace's Line and the Late Palaeozoic Gondwana–Cathaysia Divide (see **Figures 1–3**). Wallace's Line marks the boundary between Eurasian faunas and floras to the north-west (including tigers, deer, and woodpeckers) and Australasian faunas and floras to the south-east (including wallabies, possums, and cockatoos) and was recognized by and named after Alfred Russel Wallace, now regarded as the Father of Biogeography (Metcalfe et al., 2001; Van Oosterzee, 1997). The Late Palaeozoic Gondwana–Cathaysia Divide forms the boundary between high-latitude cold-climate southern-hemisphere Gondwanan faunas and floras and equatorial or northern-hemisphere warm-climate sub-tropical or tropical Cathaysian faunas and floras (Metcalfe, 1988, 1998, 2011). Both of these biogeographical boundaries are the result of convergent plate-tectonic processes bringing together allochthonous continental lithospheric terranes on which had developed contrasting faunas and

☆*Change History:* April 2013. I Metcalfe updated keywords, abstract, Further Reading, Isotopic ages, Tables 1, 2 and 3, and all figures either updated or replaced. Article text updated to reflect new data and interpretations between 2005 and 2013.

 http://dx.doi.org/10.1016/B978-0-12-409548-9.02721-4

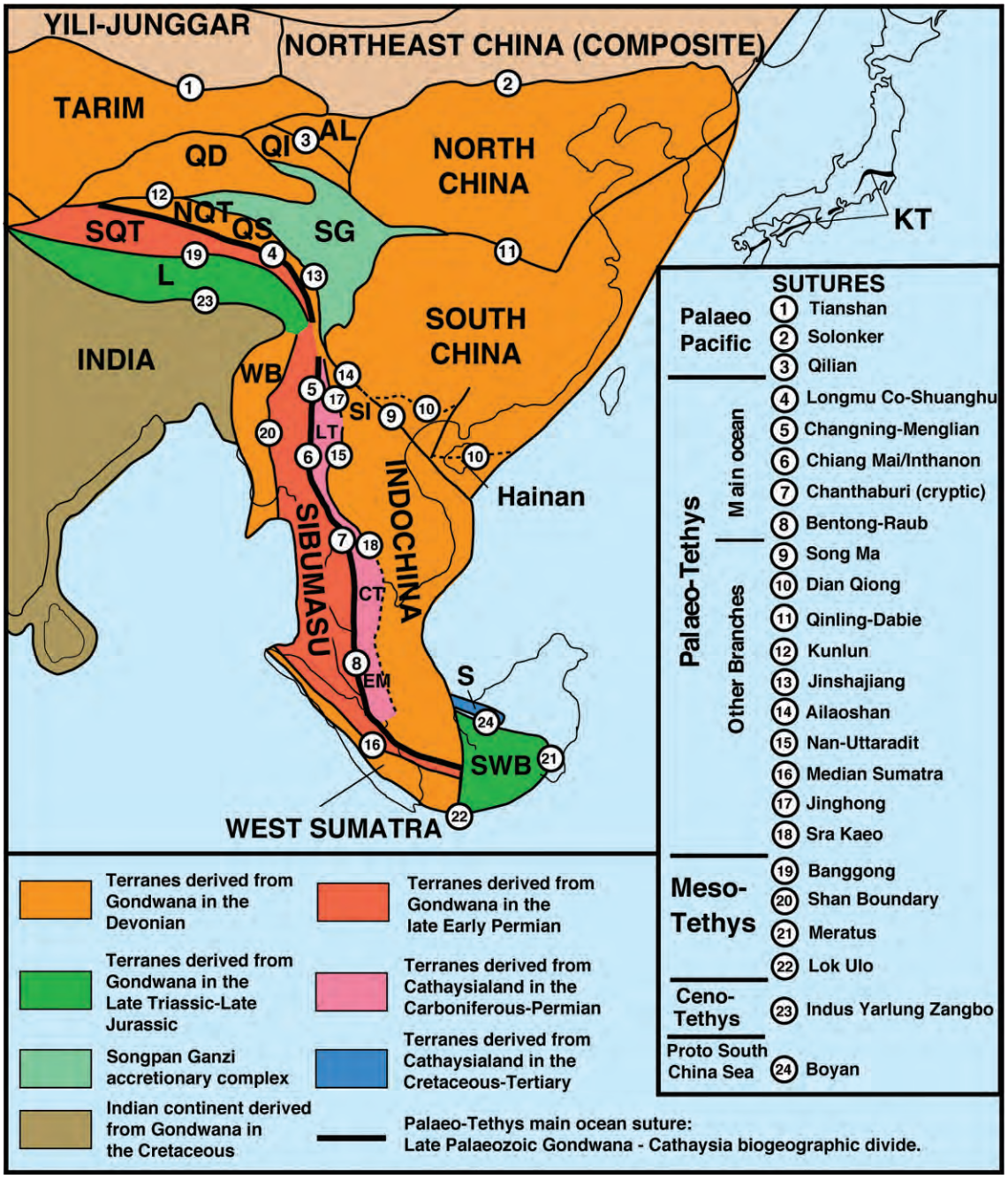


Figure 1 Distribution of principal continental blocks, arc terranes and sutures of eastern Asia. WB = West Burma, SWB = South West Borneo, S = Semitau, L = Lhasa, SQT = South Qiangtang, NQT = North Qiangtang, QS = Qamdo–Simao, SI = Simao, SG = Songpan Ganzi accretionary complex, QD = Qaidam, QI = Qilian, AL = Ala Shan, KT = Kurosegawa Terrane, LT = Lincang arc Terrane, CT = Chanthaburi arc Terrane, EM = East Malaya

floras owing to their prior geographical separation, different palaeoclimates, and biogeographical isolation. The regional geology of South-east Asia can be understood only in the context of the kinematic history and framework of the allochthonous continental terranes of the region and the evolution of the various ocean basins that once separated them (Metcalfe, 1994, 1999).

Multidisciplinary data, including stratigraphic, biostratigraphic, biogeographical, palaeoclimatological, palaeomagnetic, structural and tectonic information, indicate that in the Early Palaeozoic (541–419 Ma) all of the principal East and South-east Asian continental terranes were located on the margin of eastern Gondwana, where they formed an Indo-Australian 'Greater' Gondwana

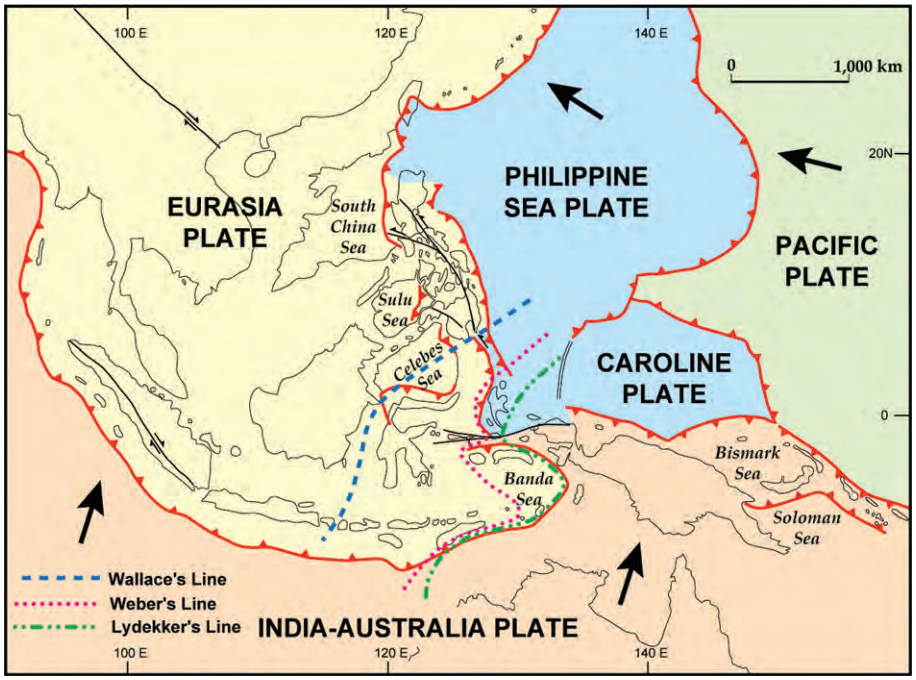


Figure 2 Principal tectonic plates of South-east Asia. Arrows show relative plate motions.

(Figure 4). They also probably formed part of the Indo-Australian element of the ancient supercontinent of Rodinia at 1000 Ma, prior to its breakup around 700 Ma and the subsequent formation of Gondwana at about 500 Ma (Figure 5).

Multidisciplinary data also suggest that the East and South-east Asian terranes rifted and separated from Gondwana as three continental slivers around 390 Ma, 290 Ma, and 200–150 Ma, in the Devonian (North China, South China, Tarim, Indochina, North Qiangtang, West Burma, West Sumatra), Lower Permian (Sibumasu, South Qiangtang), and Late Triassic–Late Jurassic (Lhasa, SW Borneo, East Java-West Sulawesi), respectively. As these three continental slivers separated from Gondwana and drifted northwards, successive ocean basins opened between each sliver and Gondwana: the Palaeo-Tethys, Meso-Tethys and Ceno-Tethys, respectively (Figure 6). The Meso-Tethys and Ceno-Tethys are broadly equivalent to the Neo-Tethys of some workers. Destruction and closure of these ocean basins by subduction during Carboniferous to Cenozoic times led to the juxtaposition, by amalgamation and accretion (continental collisions), of once widely separated continental fragments, and the remnants of the ocean basins are now preserved in the suture zones of the region (Metcalfe, 2013; Metcalfe and Allen, 2000).

Smaller continental fragments, distributed in eastern South-east Asia, were derived from Indochina and South China during the opening of the South China Sea and southwards subduction and destruction of the Proto-South China Sea, or were transported westwards along major strike-slip faults from the northern Australian margin during its collision with the westwards-moving Philippine Sea, Caroline, and Pacific plates in a kind of 'bacon-slicer' tectonic mechanism.

The regional geology of South-east Asia is thus characterized by Gondwanan dispersion and Asian accretion of terranes and the subsequent collisions of India and Australia with these terranes following the final breakup of Gondwana (Metcalfe, 1994, 1999). The geological evolution of South-east Asia is thus essentially the combined and cumulative history of these terranes, the ocean basins that once separated them, and the plate-tectonic processes that have shaped the region. A variety of multidisciplinary data (Table 1) is used to constrain the origins of the terranes, their times of rifting and separation from the parent cratons, the timing, directions, and amount of drift, and the timing of suturing (collision and welding) of the terranes to each other (Metcalfe, 1998). Some terranes sutured to each other (amalgamated) within a major ocean before, as an amalgamated composite terrane, they sutured (accreted) to proto-Asia.

Origins of the South-East Asian Terranes

Multidisciplinary data (Table 1) suggest that all the East and South-east Asian continental terranes originated on the Indian or northern or north-western Australian margin of Gondwana. Cambrian and Ordovician shallow-marine faunas of the North China, South China, and Sibumasu terranes have close affinities with those of eastern Gondwana, especially Australian Gondwana (Metcalfe, 2013). This is observed in trilobites, brachiopods, corals and stromatoporoids, nautiloids, gastropods, conodonts and

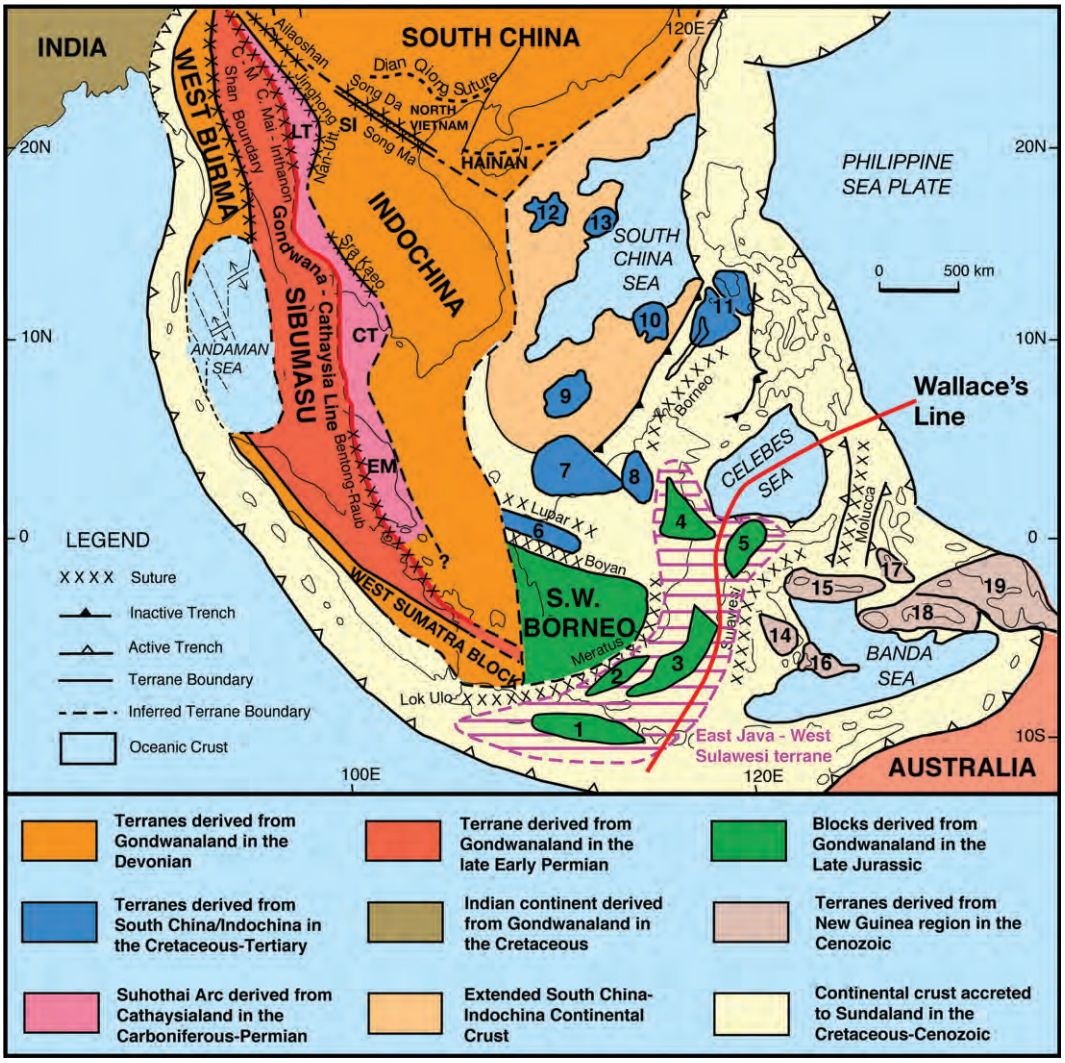


Figure 3 Distribution of continental blocks, fragments and terranes, and principal sutures of Southeast Asia. Numbered micro-continental blocks, 1. East Java 2. Bawean 3. Paternoster 4. Mangkalihat 5. West Sulawesi 6. Semitau 7. Luconia 8. Kelabit–Longbowan 9. Spratly Islands–Dangerous Ground 10. Reed Bank 11. North Palawan 12. Paracel Islands 13. Macclesfield Bank 14. East Sulawesi 15. Bangai–Sula 16. Buton 17. Obi–Bacan 18. Buru–Seram 19. West Irian Jaya. LT = Lincang Terrane, ST = Sukhothai Terrane and CT = Chanthaburi Terrane, EM = East Malaya. C–M = Changning–Menglian Suture, C.-Mai = Chiang Mai Suture, and Nan–Utt. = Nan–Uttaradit Suture.

acritarchs (**Figure 4**). The Cambrian–Ordovician faunas of Indochina are poorly known, but the Silurian brachiopods of Indochina along with those of South China, North China, Eastern Australia, and the Tarim terrane define a Sino-Australian province characterized by the *Retziella* fauna (Rong et al., 1995) (**Figures 4** and **7**). Lower Palaeozoic sequences and faunas of the Qaidam, and Ala Shan blocks are similar to those of the Tarim block and South and North China, and these blocks are regarded as disrupted fragments of a larger Tarim terrane. These biogeographical data suggest that North China, South China, Tarim (here taken to include the Qaidam and Ala Shan blocks), Sibumasu, Lhasa, South Qiangtang, North Qiangtang, Indochina, West Burma, and West Sumatra blocks formed the outer margin of northern Gondwana in the Early Palaeozoic. The close faunal affinities, at both lower and higher taxonomic levels, suggest continental contiguity of these blocks with each other and with Gondwana at this time rather than mere close proximity.

Early Palaeozoic palaeomagnetic data for the various East and South-east Asian terranes are often equivocal, varying in both quantity and quality. This makes reconstructions based purely on palaeomagnetic data difficult and suspect. However, in some cases, reasonable constraints on palaeolatitudes (but not always the hemisphere) and in some cases the actual position of attachment to Gondwana can be made (Li et al. 2004). Data from North China provide a Cambrian–Late Devonian pole-path segment that, when rotated about an Euler pole to a position of fit with the Australian Cambrian–Late Devonian pole path, positions North China adjacent to North Australia (Klootwijk, 1996). This position is consistent with reconstructions presented here. The gross stratigraphies of North China and the Arafura Basin show a remarkable similarity in the Early Palaeozoic, also supporting this position for North China. Positions for South China, Tarim, and Indochina are more equivocal, but latitudes of

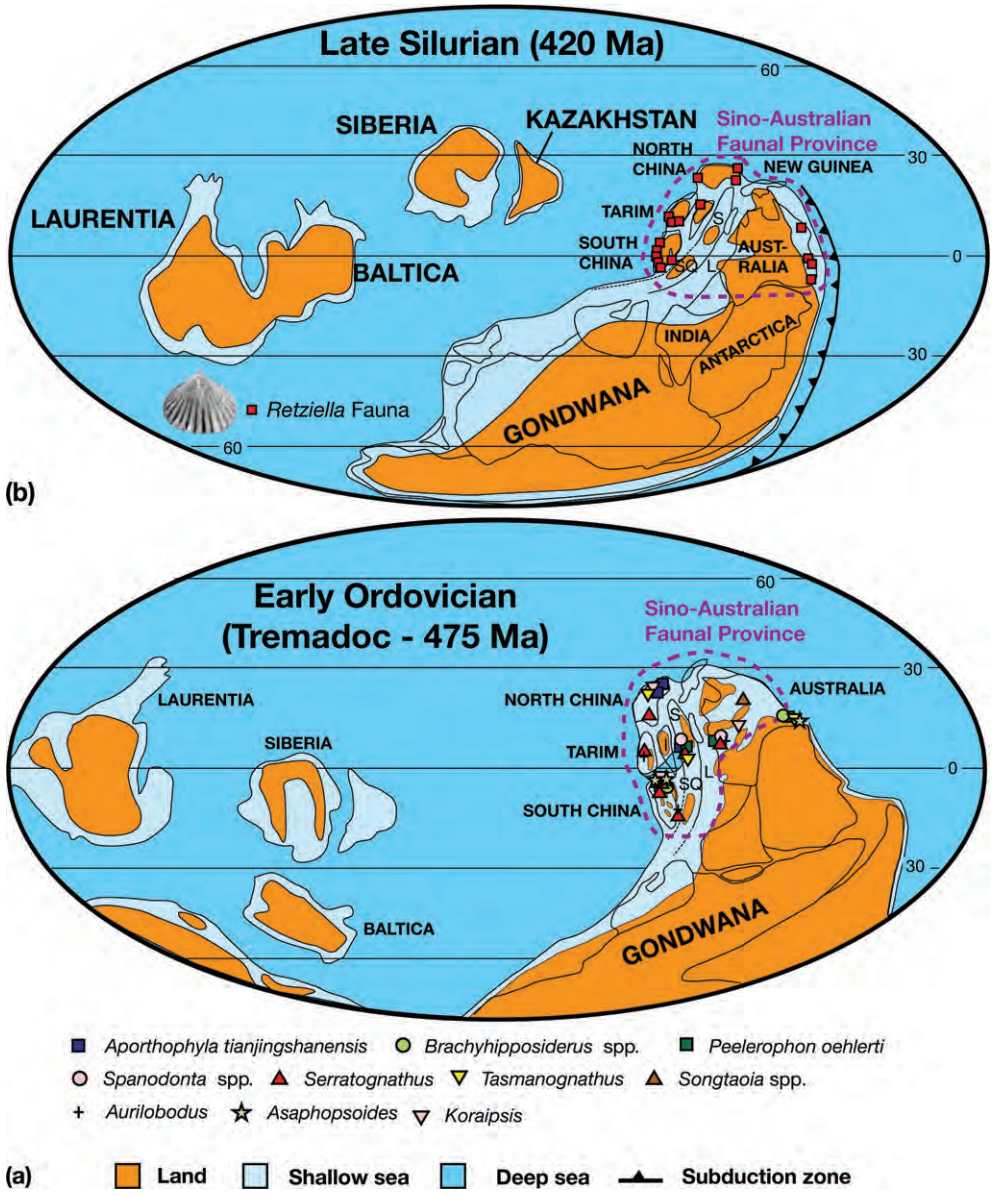


Figure 4 Palaeogeographic reconstructions for (a) Early Ordovician and (b) Late Silurian showing the postulated positions of Asian continental blocks on the Himalayan-Australian margin of Gondwana and Sino-Australian province faunas linking the Asian blocks with Australia. I = Indochina/East Malaya/West Sumatra/West Burma; SQ = South Qiangtang; L = Lhasa; S = Sibumasu.

between 1° and 15° are indicated for South China in the Late Cambrian–Early Ordovician. The Tarim Block is placed between 6°S and 20°S for the same time period, consistent with a position on the Gondwanan margin between the North and South China blocks. An origin for the Tarim Block outboard of the Kimberley region of Australia is suggested by comparisons of Precambrian sequences. The Cambrian–Early Permian faunas of the Sibumasu Terrane have unequivocal Gondwanan affinities and in particular show close relationships with western Australian faunas (Metcalfe, 2013). Gondwanan plants and spores are also reported from this terrane (Wang and Tan, 1994). Glacial–marine diamictites, with associated cold-water faunas and sediments, of possibly Late Carboniferous and (mainly) Early Permian age are also distributed along the entire length of Sibumasu (**Figure 8**) and indicate attachment to the margin of Gondwana, where substantial ice reached the sea during glaciation (Ampaiwan et al., 2009; Metcalfe, 1988; Stauffer and Lee, 1989; Stauffer and Mantajit, 1981). The most likely region for attachment of this terrane is north-western Australia, and a Late Carboniferous palaeolatitude of 42°S supports this placement. Striking similarities in the Cambrian–Early Permian gross stratigraphies of Sibumasu and the Canning Basin of north-western Australia also support a position outboard of the Canning Basin during this time (Metcalfe, 2013). Both the South Qiangtang and Lhasa blocks of Tibet exhibit Gondwanan faunas and floras up to the Early Permian and have glacial–marine diamictites, till, and associated cold-water faunas and sediments in the Late Carboniferous–Early Permian. Recent detrital zircon U-Pb and Hf isotope data from Lhasa (Zhu et al., 2011, 2012) and

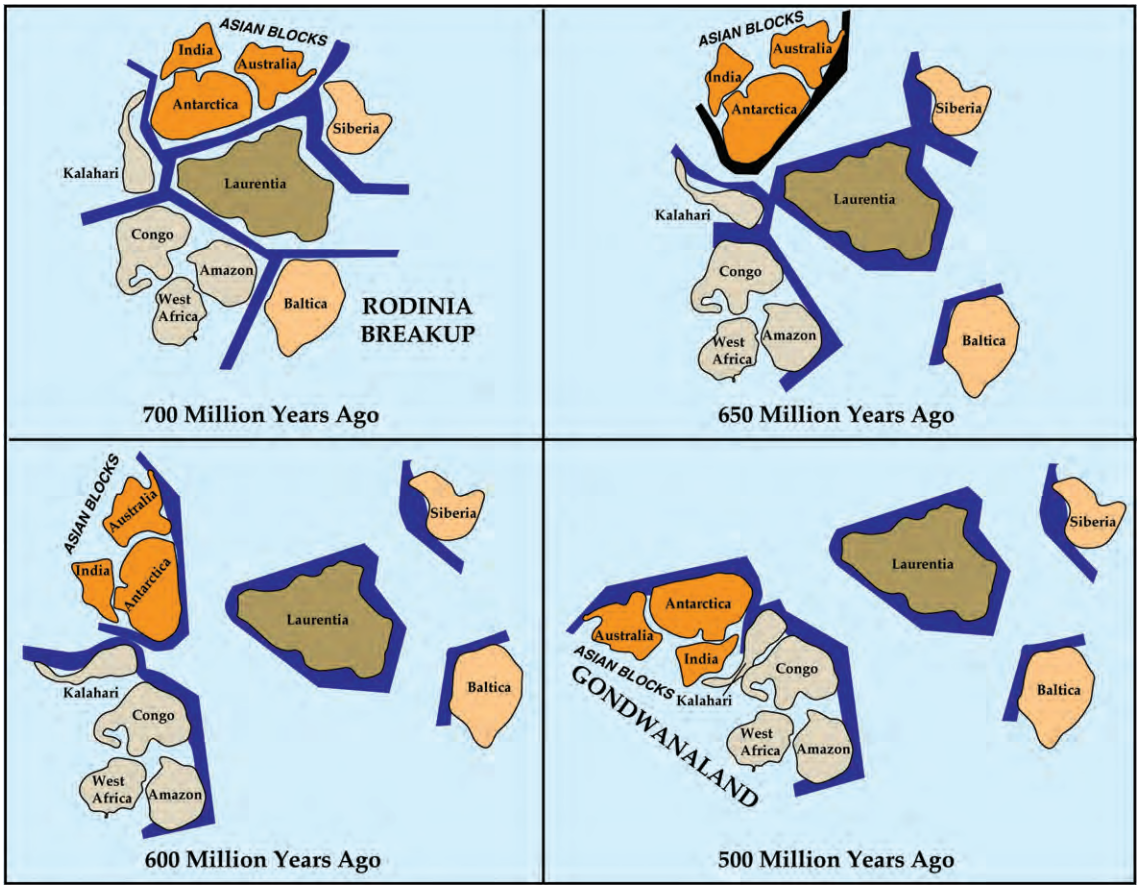


Figure 5 The supercontinent Rodinia at 700 Ma and its subsequent breakup, showing the formation of Gondwana at about 500 Ma.

Sibumasu (Hall and Sevastjanova, 2012; Sevastjanova et al., 2011) support an India-Australian Gondwana margin origin for these blocks. Thus, all the East and South-east Asian continental terranes appear to have originated on the margin of Gondwana.

Rifting and Separation of South-East Asian Terranes from Gondwana

Devonian Rifting and Separation

South China, North China, Tarim, and Indochina (including West Burma and West Sumatra) were attached to Gondwana in the Cambrian–Silurian but by Carboniferous times were separated from the parent craton, suggesting a Devonian rifting and separation of these blocks. This timing is supported by the presence of a conspicuous Devonian unconformity in South China and a subsequent Devonian–Triassic passive-margin sequence along its southern margin. Devonian basin formation in South China has also been shown to be related to rifting. The splitting of the Silurian Sino-Australian brachiopod province into two subprovinces and the apparent loss of links between the Asian terranes and Australia in the Early Devonian may be consequences of the northwards movement and separation of these terranes from Gondwana. A counterclockwise rotation of Gondwana in the Devonian about an Euler pole in Australia is also consistent with clockwise rotation of the separating Asian terranes and spreading of the Palaeo-Tethys at this time.

Carboniferous–Permian Rifting and Early Permian Separation

There is now substantial evidence for rifting along the northern margin of Gondwana in the Carboniferous–Permian accompanied by rift-related magmatism. This rifting episode led to the late Early Permian (Late Sakmarian) separation of the Sibumasu and Qiangtang terranes, as part of the Cimmerian continent, from the Indo-Australian margin of Gondwana. The late Early Permian separation and Middle–Upper Permian rapid northwards drift of the Sibumasu Terrane are supported by palaeolatitude data, which indicate a change of latitude from 42°S in the Late Carboniferous to low northern latitudes by the Early Triassic

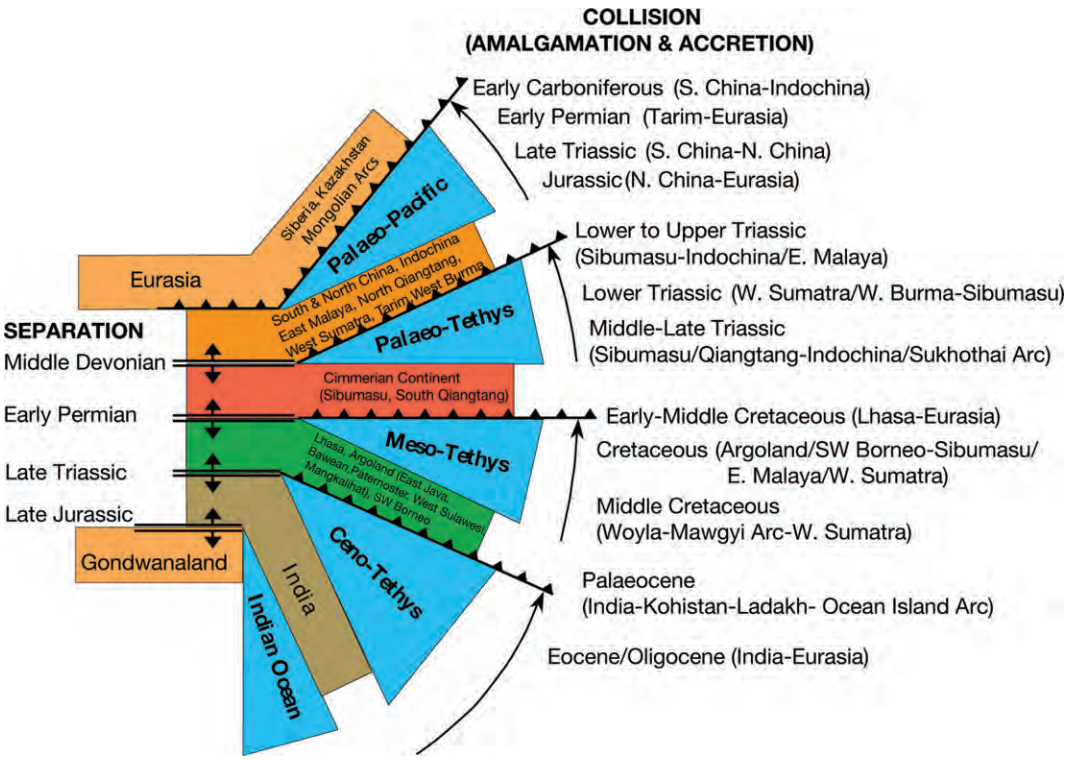


Figure 6 Schematic diagram showing times of separation and subsequent collision of the three continental slivers/collages of terranes that rifted from Gondwana and translated northwards by the opening and closing of three successive oceans, the Palaeo-Tethys, Meso-Tethys and Ceno-Tethys.

(Li et al., 2004). In addition, the faunas of the Sibumasu Terrane show a progressive change in marine provinciality from peri-Gondwanan Indoralian Province faunas in the Early Permian (Asselian–Early Sakmarian), to endemic Sibumasu province faunas in the Middle Permian, before being absorbed into the equatorial Cathaysian province in the Late Permian (Shen et al., 2013; Shi and Archbold, 1998; Ueno, 2003).

Late Triassic to Late Jurassic Rifting and Separation

Rifting, basin formation, development of unconformities, sediment source and palaeocurrent data, and ocean floor magnetic anomaly data on the NW Australian margin and in Timor (Colwell et al. 1994), indicate that a piece or pieces of continental crust rifted and separated from Australian Gondwana in the Late Triassic-Late Jurassic. These continental blocks are now identified as the Lhasa Block, East Java-West Sulawesi terrane and SW Borneo block (Hall et al., 2008, 2009; Hall, 2009; 2012; Metcalfe, 2011, 2013). Southwards subduction of the Meso-Tethys in the Triassic led to the development of a back-arc basin behind the Lhasa Block and its eventual separation from Gondwana in the Late Triassic opening the western Ceno-Tethys ocean and then the separation of SW Borneo and East Java-West Sulawesi in the Late Jurassic opening the eastern Ceno-Tethys.

Amalgamation and Accretion of Terranes

The continental terranes of East and South-east Asia progressively amalgamated during the Palaeozoic to Cenozoic. Most of the major terranes had coalesced by the end of the Cretaceous and proto South-east Asia had formed. The time of welding of one terrane to another can be determined using the various criteria given in **Table 1**. **Table 2** lists the East and South-east Asian suture zones, colliding lithospheric blocks, interpreted times of suturing, and constraints on the times of suturing. The tectonostratigraphic record of each continental terrane in the region documents the geological history of that terrane, including variations in sedimentary environment, climate, faunal and floral affinities (changes in biogeographical regime) and latitudinal shifts, rifting events, episodes of deformation, and plutono-volcanic igneous activity. The regional geological history of South-east Asia is discussed below chronologically and in terms of the evolution of the various tectonic elements of the region.

Table 1 Multidisciplinary constraining data for the origins and the rift–drift–suturing of terranes

Origin of terrane	*Age of rifting and separation*	*Drifting (palaeoposition of terrane)*	*Age of suturing (amalgamation/ accretion)*
Palaeobiogeographical constraints (fossil affinities with proposed parent craton)	Ocean floor ages and magnetic-stripe data	Palaeomagnetism (palaeolatitude, orientation)	Ages of ophiolite and ophiolite obduction ages (pre-suturing)
Tectonostratigraphic constraints (similarity of gross stratigraphy with that of parent craton, presence of distinctive lithologies characteristic of parent craton, e.g. glacials)	Divergence of apparent polar wander paths indicates separation	Palaeobiogeography (shifting from one biogeographical province to another with drift)	Melange ages (pre-suturing)
Palaeolatitude and orientation from palaeomagnetism consistent with proposed origin	Divergence of palaeolatitudes (indicates separation)	Palaeoclimatology (indicates palaeolatitudinal zone)	Age of 'stitching' plutons (post-suturing)
	Age of associated rift volcanism and intrusives		Age of collisional or post-collisional plutons (syn- to post-suturing)
	Regional unconformities (formed during pre-rift uplift and during block faulting)		Age of volcanic arc (pre-suturing)
	Major block-faulting episodes and slumping		Major changes in arc chemistry (syn-collisional)
	Palaeobiogeography (development of separate biogeographical provinces after separation)		Convergence of apparent polar wander paths
	Stratigraphy–rift sequences in grabens and half grabens		Loops or disruptions in apparent polar wander paths (indicates rapid rotations during collisions)
			Convergence of palaeolatitudes (may indicate suturing but no control on longitudinal separation)
			Age of blanketing strata (post-suturing)
			Palaeobiogeography (migration of continental animals and plants from one terrane to another indicates terranes have sutured)
			Stratigraphy/sedimentology (e.g. provenence of sedimentary detritus from one terrane on to another)
			Structural geology (age of deformation associated with collision)

Source: Reproduced with permission from Metcalfe, I. (1998). Palaeozoic and Mesozoic geological evolution of the SE Asian region: Multidisciplinary constraints and implications for biogeography. In: Hall, R., Holloway, J. D. (eds.) *Biogeography and geological evolution of SE Asia*, pp. 25–41. Amsterdam: Backhuys Publishers.

Geological and Tectonic Evolution of South-East Asia

Proterozoic (2500–541 Ma)

The East and South-east Asian terranes, together with India and Australia, formed an integral part of the ancient supercontinent of Rodinia around 1000 Ma. Fragmentation of this ancient supercontinent about 700 Ma ago led to Australia, India, Antarctica, and elements that now constitute South Africa and South America colliding and coalescing to form Gondwana about 500 Ma (**Figure 5**). Proterozoic basements of the mainland South-east Asian terranes are indicated by limited outcrops of schists and gneisses with Proterozoic radioisotopic ages in Thailand and Vietnam, by more substantial relatively unmetamorphosed Proterozoic sedimentary sequences in South China, and by Proterozoic radioisotopic ages of inherited zircons in granitoids and Proterozoic detrital zircons in Phanerozoic sedimentary rocks in the region (see Metcalfe, 2013 for details).

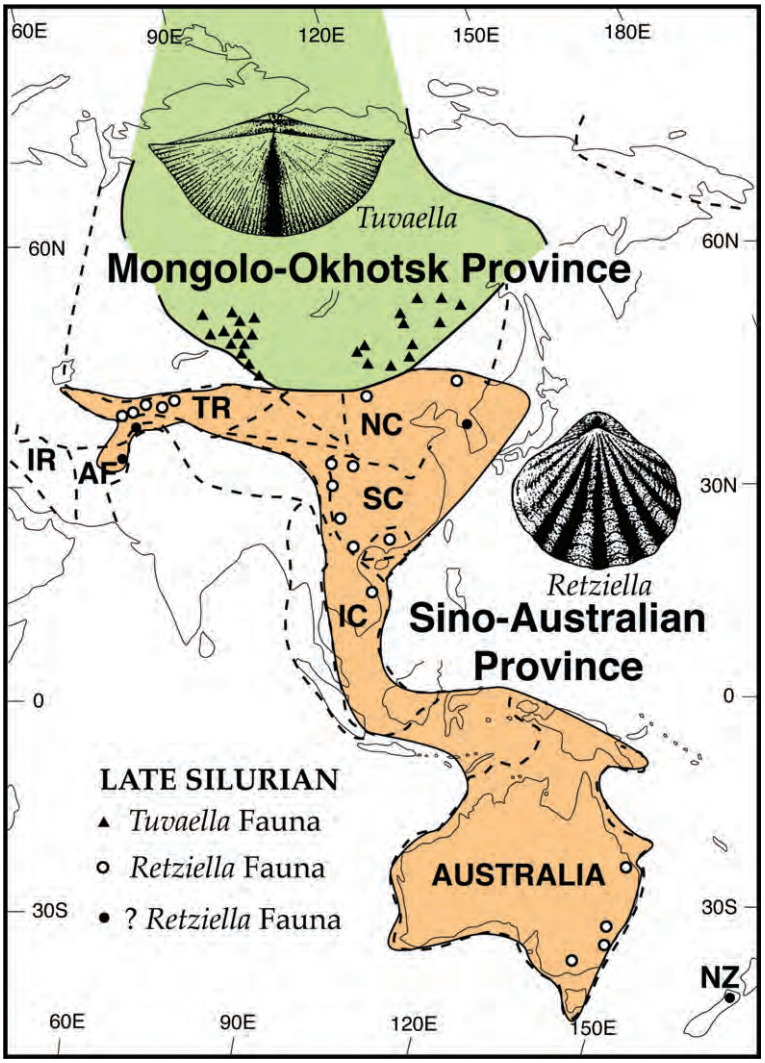


Figure 7 Late Silurian brachiopod provinces of Asia–Australasia. IR, Iran; AF, Afghanistan; TR, Tarim; NC, North China; SC, South China; IC, Indochina; NZ, New Zealand.

Phanerozoic (541–0 Ma)

Table 3 summarizes the principal geological events, and their timings, that have affected the South-east Asian region during the Palaeozoic, Mesozoic, and Cenozoic Eras. Continental collisions (amalgamation or accretion) are dated by the various suture zones (Table 2). The principal events that have affected the South-east Asian region during the Phanerozoic are outlined below chronologically.

Cambrian–Ordovician–Silurian (541–419 Ma)

The East and South-east Asian terranes formed part of Indian–Australian 'Greater Gondwana'. Faunas of this age on the Asian blocks and in Australasia define Asian–Australian palaeo-equatorial warm-climate 'provinces', for example the Sino-Australian brachiopod province in the Silurian – see Figures 4 and 7.

Devonian (419–359 Ma)

Australian eastern Gondwana continued to reside in low southern latitudes during the Devonian but rotated counterclockwise. This counterclockwise rotation mirrors a clockwise rotation of the North and South China, Tarim and Indochina terranes as they separated from Gondwana as an elongate continental sliver. Separation of this sliver from Gondwana opened the Palaeo-Tethys Ocean (Figure 9). Devonian faunas on the Chinese terranes still have some Australian connections. Early Devonian endemic faunas of South China, including some fish faunas and the distinctive *Chuiella* brachiopod fauna (Figure 9), are interpreted to be a

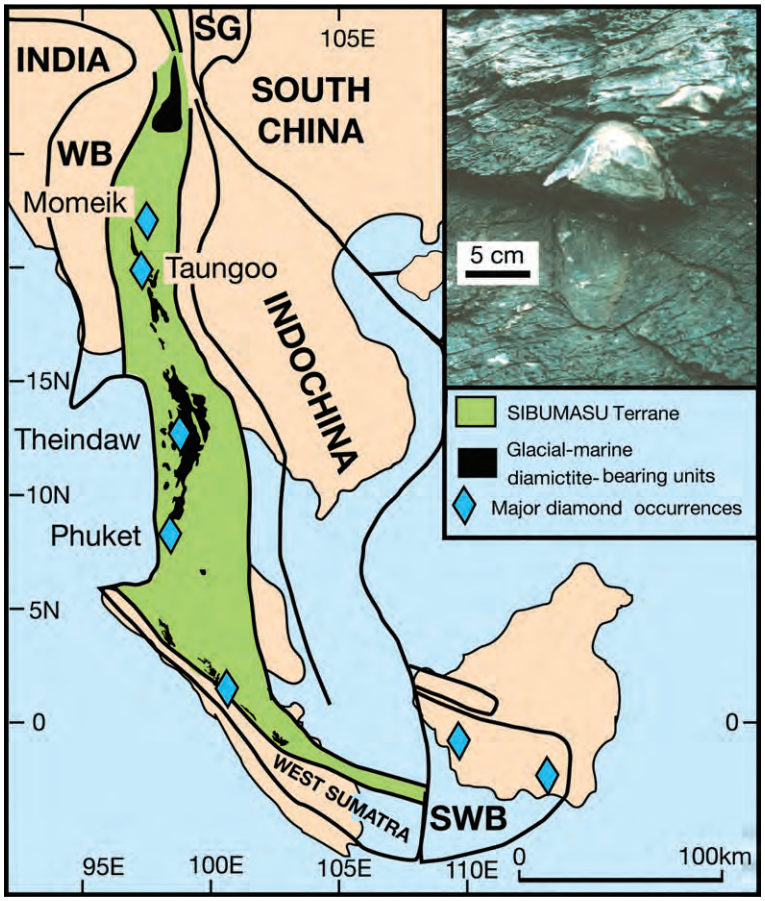


Figure 8 Map of mainland South-east Asia, showing the distributions of Carboniferous–Early Permian glacial–marine sediments and major alluvial diamond deposits. Inset photo: dropstone in glacial–marine diamictite orientated vertical to bedding; Singa Formation, Langkawi Islands, Peninsular Malaysia.

result of the rifting process and isolation of South China on the rifting continental promontory, and do not necessarily imply continental separation of South China from the other Chinese blocks and Australia at this time.

Carboniferous (359–299 Ma)

During the Carboniferous, Gondwana rotated clockwise and collided with Laurentia in the west to form Pangaea. Australia, Sibumasu, South Qiangtang and Lhasa were still attached to north-eastern Gondwana and drifted from low southern latitudes in Tournaisian–Visean times to high southern latitudes in late Serpukhovian-Bashkirian times. The major phase of Gondwanan glaciation with widespread ice sheets commenced in the Serpukhovian and extended through to the Early Permian (Fielding et al., 2008). There were major global shifts in both plate configurations and climate during this time, and there was a change from warm to cold conditions in Australasia. This is reflected in the change from a high to a low diversity of faunas in Australasia, especially eastern Australia, where low-diversity endemic faunas developed in the Late Carboniferous. The faunas and floras of North and South China, Indochina, East Malaya, West Sumatra, and Tarim are tropical or sub-tropical Cathaysian or Tethyan types during the Carboniferous and show no Gondwanan affinities (**Figure 10**). These terranes had already separated from Gondwana and were located in low-latitude or equatorial positions during the Carboniferous (**Figure 9**). Indochina and South China collided and amalgamated within Tethys during the Early Carboniferous along the Song Ma suture zone, which is now located in Laos and Vietnam. Ice-sheets and glaciers extended across much of eastern Gondwana during the Late Carboniferous: ice reached the marine environment of the India–Australian continental shelf of Gondwana, and glacial–marine sediments (diamictites; pebbly mudstones interbedded with normal marine shales and sands) were deposited on the continental shelf of eastern Gondwana.

Early Permian (299–272 Ma)

During the Permian, Australia remained in high southern latitudes. Glacial ice continued to reach the marine environment of the north-eastern Gondwanan margin, and glacial–marine sediments continued to be deposited on the Sibumasu, Qiangtang, and

Lhasa terranes (Figures 8, 11 and 12). Gondwanan cold-climate faunas and floras characterized the Sibumasu, South Qiangtang, and Lhasa terranes at this time. In addition, the distinctive cool-water-tolerant conodont genus *Vjalovognathus* defines an eastern peri-Gondwanan cold-water province (Figure 11). Floral provinces are particularly marked at this time, and the distinctive Cathaysian (*Gigantopteris*) flora developed on North China, South China and Indochina which were isolated within Tethys and located equatorially (Figure 13). During the late Early Permian (end-Sakmarian), the Cimmerian continental sliver separated from the north-eastern margin of Gondwana, and the Meso-Tethys Ocean opened by seafloor spreading between it and mainland

Table 2 East and South-east Asian sutures and their interpreted ages and age constraints. For location of sutures see Figure 1 and Figure 3 and Metcalfe, 2013 for further details

No. on Fig. 1	*Suture name*	*Colliding lithospheric blocks*	*Suture age*	*Age constraints*
1	Tianshan	Tarim, Kazakhstan	Permian	Lower Carboniferous ophiolites. Major arc magmatism ceased in the Late Carboniferous. Late Permian post-orogenic subsidence and continental sedimentation in Junggar Basin. Palaeomagnetic data indicate convergence of Tarim and Kazakhstan by the Permian. Upper Permian continental clastics blanket the suture.
2	Solonker	North China, Altaid terranes	Jurassic	Late Jurassic–Early Cretaceous deformation and thrust faulting. Widespread Jurassic–Cretaceous granites. Triassic–Middle Jurassic deep-marine cherts and clastics. Upper Jurassic–Lower Cretaceous continental deposits blanket suture.
3	Qilian	Ala Shan, Qilian	Ordovician	Andean-type active continental margin represented by infant arc magmatism of ~517–490 Ma. Ophiolite formed in a back-arc basin setting between ~490–445 Ma.
4	Longmu Co-Shuanghu	South Qiangtang, North Qiangtang-Qamdo-Simao	Late Triassic	Suture zone rocks include Devonian and Carboniferous turbiditic 'flysch' and Late Devonian and Permian pelagic radiolarian cherts. Ar/Ar ages on phengite between 203–222 Ma and Lu/Hf isochron ages on eclogite range between 233–244 Ma. Ocean-floor basic extrusives are of Permian age and Carboniferous–Permian mélange is reported. Carboniferous–Permian island arc rocks are developed along the west side of the suture. Upper Triassic collisional granitoids are associated with the suture.
5	Changning-Menglian	Sibumasu, Sukhothai Arc	Late Triassic	Oceanic ribbon-bedded chert–shale sequences have yielded graptolites, conodonts, and radiolarians indicating ages ranging from Lower Devonian to Middle Triassic. Limestone blocks and lenses dominantly found within the basalt sequence of the suture and interpreted as seamount caps, have yielded fusulinids indicative of Lower Carboniferous to Late Middle Permian ages. Supra-subduction zone ophiolites are dated at ca. 270–264 Ma (Permian). Other ophiolites that appear to represent initial ocean floor spreading are dated at 386 Ma (Early-Middle Devonian boundary).
6	Chiang Mai-Inthanon	Sibumasu, Sukhothai Arc	Late Triassic	Basic volcanics (including pillow basalts) are dated as Carboniferous and Permian. Ages of oceanic deep-marine bedded cherts within the suture zone range from Middle Devonian to Middle Triassic. Seamount limestone caps are dated as Lower Carboniferous (Visean) to Upper Permian (Changxingian) in age
7	Chanthaburi (Klaeng Tectonic Line)	Sibumasu, Sukhothai Arc	Late Triassic	Late Devonian, Late Permian and Middle Triassic radiolarian cherts.

(*Continued*)

Table 2 (Continued)

No. on Fig. 1	*Suture name*	*Colliding lithospheric blocks*	*Suture age*	*Age constraints*
8	Bentong-Raub	Sibumasu, Sukhothai Arc (East Malaya)	Middle Triassic	Upper Devonian to Upper Permian oceanic ribbon-bedded cherts. Mélange includes chert and limestone clasts with Lower Carboniferous to Upper Permian ages. Schists and phyllites representing Sibumasu continental margin-slope deposits incorporated into the accretionary complex are dated as Ordovician, Silurian and Devonian. The Main Range 'collisional' 'S' type granites of Peninsular Malaysia range from Late Triassic (230 ± 9 Ma) to earliest Jurassic (207 ± 14 Ma) in age, with a peak of around 210 Ma. Suture zone is covered by latest Triassic, Jurassic, and Cretaceous, mainly continental, blanketing sequence.
9	Song Ma	Indochina, South China	Disputed: Variously interpreted as Late Devonian, Early Carboniferous, Late Permian–Early Triassic, and Middle Triassic	Sm/Nd isochron dating of titanites from serpentinites (387–313 Ma) indicate Middle Devonian-Carboniferous crystalization ages. Large-scale folding and thrusting and nappe formation in the Early to Middle Carboniferous. Middle Carboniferous shallow-marine carbonates are reported to blanket the Song Ma suture in North Vietnam. Pre-middle Carboniferous faunas on each side of the Song Ma zone are different whilst the Middle Carboniferous faunas are essentially similar. Carboniferous floras on the Indochina block in north-eastern Thailand indicate continental connection between Indochina and South China in the Carboniferous. Granitoids along the Trung Song belt are dated as Middle Permian-Early Triassic. A U-Pb SHRIMP zircon age of 230 ± 8.2 Ma (early Late Triassic) for the Song Ma eclogites is interpreted to represent the closure age of the Palaeo-Tethys along the Song Ma suture. See Metcalfe, 2013 for full discussion.
10	Dian Qiong	South China, Indochina	Triassic	Devonian-Permian radiolarian cherts with associated mid-ocean ridge basalts.
11	Qinling-Dabie	South China, North China	Triassic–Jurassic	Late Triassic subduction-related granite. Late Triassic U–Pb dates of zircons from ultrahigh-pressure eclogites. Late Triassic–Early Jurassic convergence of apparent polar wander-paths and palaeolatitudes of South China and North China. Initial contact between South and North China is indicated by isotopic data in Shandong and sedimentological records along the suture. Widespread Triassic to Early Jurassic deformation in the North China block north of the suture
12	Kunlun	North Qiangtang, Qaidam	Triassic	Permo-Triassic ophiolites and subduction-zone mélange. Upper Permian calc-alkaline volcanics, strongly deformed Triassic flysch and Late Triassic granitoids
13	Jinshajiang	North Qiangtang, South China	Late Triassic	Ophiolites are regarded as Upper Permian to Lower Triassic in age. Mélange comprises Devonian, Carboniferous, and Permian exotics in a Triassic matrix. Upper Permian to Jurassic sediments unconformably overlie Lower Permian ophiolites in the Hoh Xil Range

Table 2 (Continued)

No. on Fig. 1	*Suture name*	*Colliding lithospheric blocks*	*Suture age*	*Age constraints*
14	Ailaoshan Suture	Simao, South China	Middle Triassic	NMORB ophiolites dated as Middle-Late Devonian (c. 387–374 Ma). EMORB ophiolites dated as Lower Carboniferous (c. 346–341 Ma). Plagiogranite (Shuanggou ophiolite) has been dated as latest Devonian (c. 362 Ma). Ophiolitic rocks are associated with deep-marine sedimentary rocks including ribbon-bedded cherts that have yielded some Lower Carboniferous and Lower Permian radiolarians. Upper Triassic sediments (Carnian conglomerates and sandstones, Norian limestones, and Rhaetian sandstones) blanket the suture
15	Nan-Uttaradit Suture	Sukhothai Arc, Indochina	Late Triassic	Bedded radiolarian charts are Permian-Middle Triassic age. Ophiolitic mafic and ultramafic rocks with associated blueschists are dated as Permian. Mafic and ultramafic blocks in the mélange comprise ocean-island basalts, back-arc basin basalts and andesites, island-arc basalts and andesites and supra-subduction cumulates generated in Carboniferous to Permo-Triassic times. Permo-Triassic dacites and rhyolites associated with relatively unmetamorphosed Lower Triassic sandstone–shale turbidite sequence. Suture zone rocks are overlain unconformably by Jurassic redbeds and post-Triassic intraplate continental basalts
16	Median Sumatra	West Burma, Sibumasu	Early Triassic	No remnants of the ocean basin that once separated West Sumatra and Sibumasu so far known. It is likely that West Sumatra was slid into juxtaposition with Sibumasu from east to west along a major strike-slip fault associated with oblique subduction
17	Jinghong	Sukhothai Arc, Indochina	Late Triassic	Sture zone rocks include melange, serpentinites tholeiitic basalts and cherts. Deep-marine radiolarian cherts are of late Early, Middle and Late Permian age.
18	Sra Kaeo Suture	Sibumasu, Indochina	Late Triassic	Suture zone rocks include melange and chert–clastic sequences, which include ultrabasics, serpentinites, Carboniferous pillow basalts, and Early Permian to Late Triassic oceanic sediments and associated pillow basalts. Limestone blocks in the mélange range from Upper Lower Permian to Middle Permian and a granitic lens has yielded a zircon U–Pb age of 486 ± 5 Ma. Imbricate thrust slices dated as Middle Triassic by radiolarians. Jurassic continental sandstones blanket the suture zone
19	Banggong-Nujiang Suture	Lhasa, South Qiangtang	Late Jurassic–Earliest Cretaceous	Ophiolites along the suture zone are dated as Late Triassic-Early Jurassic. Serpentinite matrix melange contains blocks of peridotite, mafic lavas and dikes, and amphibolite blocks of both arc and MORB affinity of Early-Middle Jurassic ages. Suture is blanketed in Tibet by Cretaceous and Palaeogene rocks. Structural data indicate collision around the Jurassic–Cretaceous boundary
20	Shan Boundary Suture	West Burma, Sibumasu	Early Cretaceous	Cretaceous thrusts in the back-arc belt. Late Cretaceous age for the Western Belt tin-bearing granites

(*Continued*)

Table 2 (Continued)

No. on Fig. 1	*Suture name*	*Colliding lithospheric blocks*	*Suture age*	*Age constraints*
21	Meratus Suture	East Java-West sulawesi, SW Borneo	Late Cretaceous	Bedded radiolarian cherts range in age from early Middle Jurassic to late Early Cretaceous. Subduction mélange of Middle–Late Cretaceous age. Ophiolite with Jurassic ultramafic rocks. Pillow basalts of Jurassic and Early Cretaceous ages. Turbiditic flysche of Early–Late Cretaceous age. Ophiolite obducted in Cenomanian. Suture overlain by Eocene strata.
22	Lok Ulo	East Java-West sulawesi, SW Borneo	Late Cretaceous	Bedded radiolarian cherts range in age from Early-Late Cretaceous. Subduction mélange of Late Cretaceous age. Ophiolite with Late Jurassic ultramafic rocks. Pillow basalts of latest Jurassic and Earliest Cretaceous ages. Suture overlain by Eocene strata
23	Indus-Yarlung-Zangbo Suture	India, Eurasia	Disputed: Late Cretaceous–Eocene	The suture zone includes what has been interpreted as intra-oceanic arcs, ophiolites and accretionary complexes formed within the Ceno-Tethys. Late Permian Changhsingian seamount limestones occur in the suture north of Burang. Jurassic–Cretaceous ophiolites and ophiolitic mélange with Jurassic–Lower Cretaceous radiolarian cherts. Late Cretaceous blueschists. Eocene collision-related plutons. Palaeomagnetic data indicates initial collision around 60 Ma but this is interpreted as collision with and oceanic arc by some authors. Final hard collision may be as young as c 35 Ma.
24	Boyan Suture	Semitau, SW Borneo	Late Cretaceous	Upper Cretaceous mélange

Gondwana (**Figure 11(c)**). Subduction beneath South China–Indochina in the Early Permian led to the development of the Sukhothai Arc on the margin of Indochina and development of the short lived Sukhothai back-arc basin represented by the Jinghong, Nan-Uttaradit and Sra Kaeo suture zones (Metcalfe, 2011, 2013; Sone and Metcalfe, 2008).

Late Permian (272–252 Ma)

By early Late Permian times the Sibumasu and Qiangtang terranes had separated from Gondwana, and the Meso-Tethys had opened between this continental sliver and Gondwana (**Figure 11(c)**). The Palaeo-Tethys continued to be destroyed by northwards subduction beneath Laurasia, North China, and the amalgamated South China–Indochina terranes. Following separation, and during their northwards drift, the Sibumasu and South Qiangtang terranes initially developed a Cimmerian Province fauna and were then absorbed into the Cathaysian Province. North and South China begin to collide during the Late Permian, and a connection between mainland Pangaea and Indochina, via South and North China or via the western Cimmerian continent, is indicated by the occurrence of the genus *Dicynodon* in the Upper Permian of Indochina (Battail, 2009) (**Figure 11(c)**). Sibumasu began to collide with the Sukhothai Arc in the Late Permian, and collision with the arc and with Indochina continued into the Triassic (Metcalfe, 2011, 2013). Deformation associated with this event, and with the collision of North and south China, is known as the Indosinian Orogeny in South-east Asia and China.

Triassic (252–201 Ma)

Australia was in low-to-moderate southern latitudes during the Triassic. The Sibumasu and South Qiangtang terranes collided and sutured to the Indochina–South China amalgamated terrane. The West Burma and West Sumatra blocks were pushed westwards by the interaction of the westwards-subducting Palaeopacific Plate with the northwards-subducting Palaeo-Tethys during the Sibumasu–Sukhothai Arc-Indochina collisional process and was translated along major strike-slip faults to a position outboard of Sibumasu in the Early Triassic. The Sukhothai back-arc basin was closed in the Middle to early Late Triassic. The North–South China collision was nearly complete, and ultrahigh-pressure metamorphics were exhumed along the Qinling-Dabie suture zone. Sediment derived from the North–South China collisional orogen poured into the Songpan Ganzi accretionary-complex basin

Table 3 Palaeozoic, Mesozoic, and Cenozoic events and their ages in South-east Asia

Process	*Age*
Palaeozoic evolution	
Rifting of South China, North China, Indochina, North Qiangtang, East Malaya, West Sumatra, West Burma, Tarim, and Qaidam from Gondwana	Early Devonian
Initial spreading of the Palaeo-Tethys ocean	Middle–Late Devonian
Amalgamation of South China and Indochina.	Late Devonian to Early Carboniferous
Development of the Jinghong-Nan-Uttaradit-Sra Kaeo back-arc basin and separation of the Sukhothai Arc by back-arc spreading	Early-Middle Permian
Rifting of Sibumasu and South Qiangtang from Gondwana as part of the Cimmerian continent	Late Early Permian (Sakmarian)
Initial spreading of Meso-Tethys ocean	Middle Permian
Collision and suturing of Sibumasu to Sukhothai Arc and Indochina.	Latest Permian to Middle Triassic
Initial collision of South and North China and development of Tanlu Fault	Late Permian to Triassic
Emplacement of West Sumatra and West Burma outboard of Sibumasu by strike-slip tectonics.	Triassic
Mesozoic evolution	
Suturing of South China with North China and final consolidation of Sundaland	Late Triassic to Early Jurassic
Rifting of Lhasa, East Java-West Sulawesi and SW Borneo	Late Triassic to Late Jurassic
Initial spreading of Ceno-Tethys ocean	Late Triassic (Norian) in west (northern India) and Late Jurassic in east (NW Australia)
Northward drift of Lhasa, East Java-West Sulawesi and SW Borneo	Jurassic to Cretaceous
Collision of the Lhasa Block with Eurasia	Late Jurassic–Earliest Cretaceous
Accretion of East Java-West Sulawesi to SW Borneo	Late Cretaceous
Suturing of Semitau to SW Borneo	Late Cretaceous
Cenozoic evolution	
Collision of India with Kohistan-Ladakh Arc	c 60 Ma, Palaeocene
Collision of India with Eurasia	Eocene but precise age debated between 45 and 35 Ma
Final separation of Australia from Antarctica and establishment of the circum-Antarctic ocean current	45 Ma, Late Eocene
Northwards drift of the isolated Australian continent over 27° of latitude. Gradual drying of the continent and evolution of the distinctive Australian fauna and flora from Gondwana ancestry	45–0 Ma, Late Eocene to present
Clockwise rotation and extrusion of Indochina and parts of northern Sibumasu	During the early Cenozoic but precise age not known
Major strike-slip faulting	30–15 Ma, Middle Oligocene to Middle Miocene
Anti-clockwise rotation of Borneo and the Malay Peninsula	Progressively during the Cenozoic but mostly during the Miocene
Initial spreading of the South China Sea basin	30 Ma, Middle Oligocene
Proto-South China Sea destroyed by southwards subduction	40–15 Ma, Middle Eocene to Middle Miocene
Collision of the Australian continent with the Philippine Sea Plate (initiating clockwise rotation of Philippine Sea Plate and causing structural inversion in many Cenozoic basins of the region)	25 Ma, Late Oligocene
Clockwise rotation of the Philippine Sea Plate	From about 20 Ma onwards
Opening of the Sulu Sea	20 Ma, Early Miocene
Molucca Sea double subduction established	10 Ma, Late Miocene
Collision of the Philippine Arc and Eurasian continental margin in Taiwan	5 Ma, End Miocene

producing huge thicknesses of flysch turbidites. By Late Triassic (Norian) times, North China, South China, Sibumasu, Sukhothai Arc, Indochina, West Burma and West Sumatra had coalesced to form proto-East and South-east Asia (**Figure 11(d)**). During the final collisional consolidation of these terranes, the major economically important Late Triassic–Early Jurassic collisional tin-bearing Main Range granitoids were formed in South-east Asia (**Figure 14**).

Jurassic (201–145 Ma)

Australia remained in low-to-moderate southern latitudes in the Jurassic. Rifting and separation of the Lhasa, East Java-West Sulawesi and SW Borneo terranes from north-western Australia occurred progressively from west to east during the Late Triassic to Late Jurassic. The Ceno-Tethys Ocean opened behind these terranes as they separated from Gondwana (**Figure 15**). Final welding of North China to Eurasia (the Yanshanian deformational orogeny) took place in the Jurassic with the closure of the Mongol-Okhotsk Ocean. Initial pre-breakup rifting of the main Gondwanan supercontinent also began in the Jurassic.

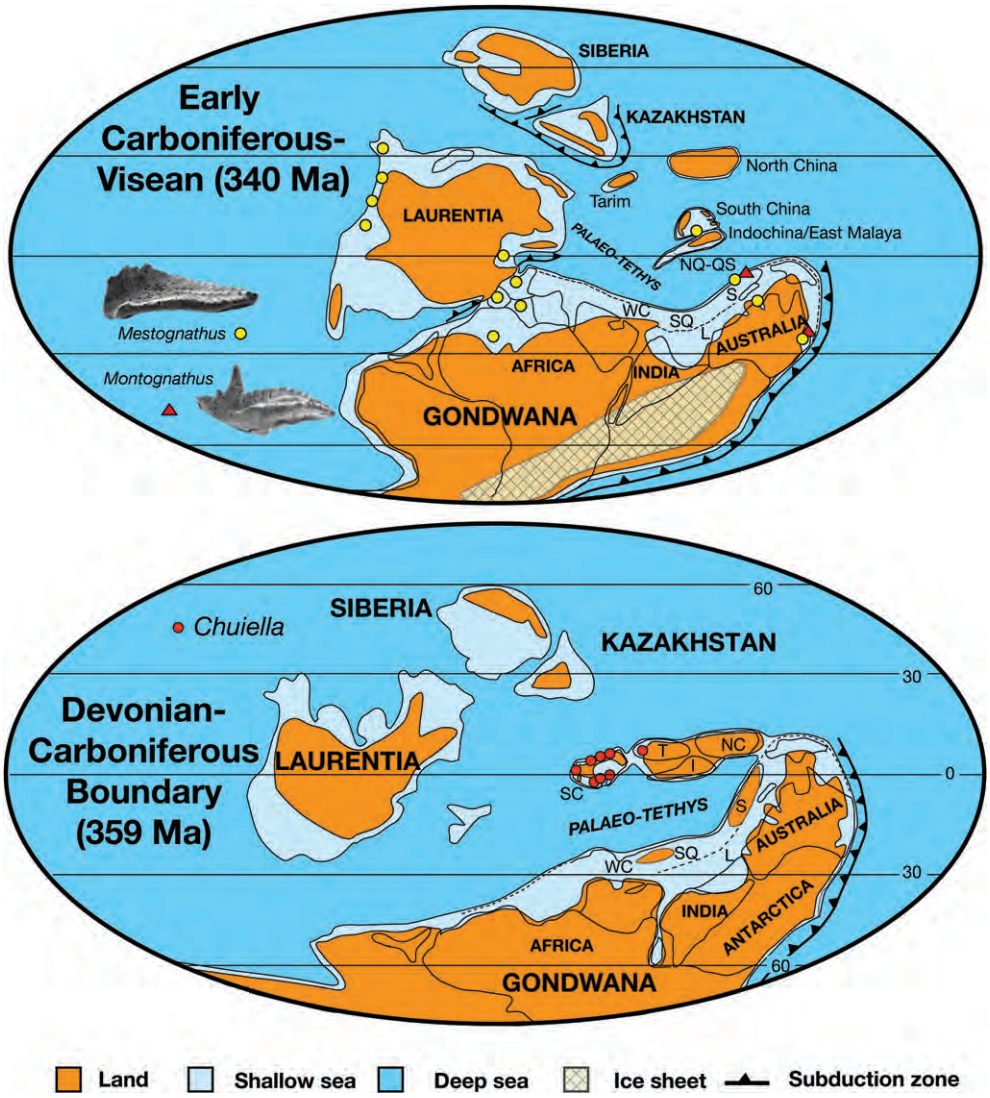


Figure 9 Reconstructions of eastern Gondwana at (a) Devonian-Carboniferous boundary and (b) Early Carboniferous (Visean) times showing the postulated positions of the East and Southeast Asian terranes. Also shown is the distribution of the endemic Tournaisian brachiopod genus Chuiella and the biogeographic distributions of the conodont genera *Mestognathus* (Illustrated specimen is *Mestognathus beckmanni* from the Kanthan Limestone, Peninsular Malaysia) and *Montognathus* (*Montognathus carinatus* from Peninsular Malaysia illustrated). NC = North China; SC = South China; T = Tarim; I = Indochina/East Malaya/West Sumatra/West Burma; SQ = South Qiangtang; NQ-QS = North Qiangtang-Qamdo-Simao; L = Lhasa; S = Sibumasu; and WC = Western Cimmerian Continent.

Cretaceous (145–66 Ma)

The Lhasa Block collided and amalgamated with Eurasia in the Cretaceous closing the western Meso-Tethys ocean. Gondwana broke up and India drifted north, making initial contact with an intra oceanic arc at about 55 Ma (interpreted here as the Kohistan-Ladakh-Woyla arc following Metcalfe, 2013) (**Figure 15**). The Meso-Tethys north of the East Java-West Sulawesi and SW Borneo blocks closed in the Late Cretaceous when these blocks collided with Sundaland. Australia began to separate from Antarctica and drift northwards, but a connection with Antarctica via Tasmania remained.

Cenozoic (66–0 Ma)

The Cenozoic evolution of East and South-east Asia involved substantial movements along and rotations of strike-slip faults, rotations of continental blocks and oceanic plates, the development and spreading of 'marginal' seas, and the formation of important hydrocarbon-bearing sedimentary basins. This evolution was essentially due to the combined effects of the interactions of the Eurasian, Pacific, and Indo-Australian plates and the collisions of India with Eurasia and of Australia with South-east Asia.

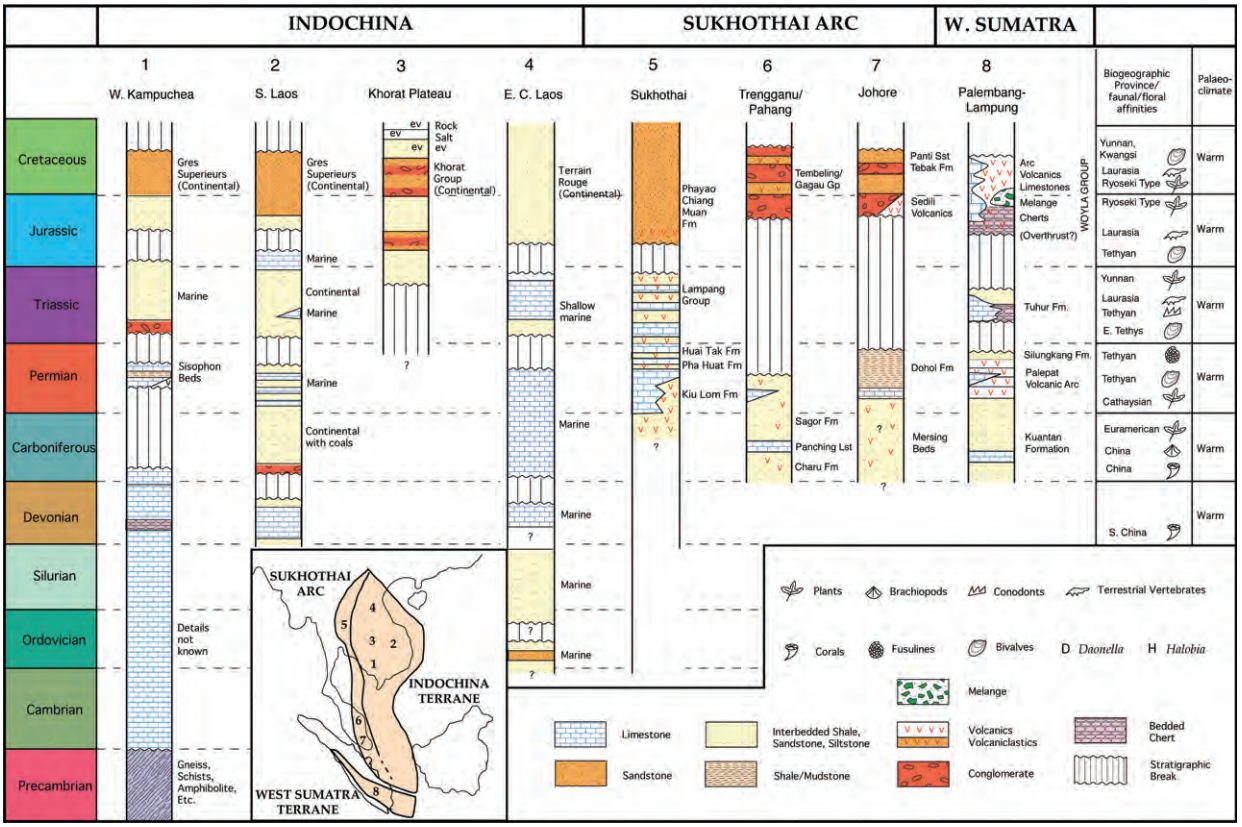


Figure 10 Representative stratigraphic columns for the Indochina Block, Sukhothai Arc, and West Sumatra Block, and faunal provinces and affinities of faunas and floras. ev = evaporites.

Various tectonic models have been proposed for the Cenozoic evolution of the region, which invoke different mechanisms for the India–Eurasia collision and different interpretations of rotations of continental blocks and oceanic plates. Three basic mechanisms have been proposed for the northwards motion of India into Eurasia. The first involves underthrusting of greater India beneath Eurasia (Powell and Conaghan, 1973; Zhao et al., 1993; Zhou and Murphy, 2005), the second involves crustal shortening and thickening (Chang et al., 1986; Dewey and Burke, 1973; Dewey et al., 1988); and the third involves major eastwards lateral extrusion with large displacements along strike-slip faults and progressive clockwise rotations of China, Indochina, and Sundaland (Tapponnier et al., 1982, 1986). The timing of collision between India and Eurasia is also still hotly debated with estimates ranging from 60 Ma (Yin, 2010) to 35 Ma (Ali and Aitchison, 2007, 2008). Proponents for a younger collision suggest that the collision at c 60–55 Ma was between India and an intra-oceanic island arc, here identified as the Kohistan-Ladakh-Woyla arc.

Cenozoic clockwise rotations of crustal blocks are observed in Indochina and western Thailand, but major progressive counterclockwise rotations are seen in Borneo and the Malay Peninsula. The counterclockwise rotations observed in Borneo and Malaya, and less than expected displacements along major strike slip faults are at variance with the extrusion model. Most models for the region neglect the clockwise rotation of the Philippine Sea Plate and/or the counterclockwise rotation of Borneo. The Cenozoic reconstruction model proposed by Hall, 2002, 2012 and illustrated here (**Figures 16–18**), which takes into account both the clockwise rotation of the Philippine Sea Plate and the counterclockwise rotation of Borneo and Peninsular Malaysia, is preferred.

Knowledge of the Cenozoic geodynamics of South-east Asia, the movements of microcontinents, and the shifting distribution of land and sea in the area (**Figure 19**) underpins and enhances our understanding of the biogeography of the region (Hall and Holloway, 1998; Metcalfe, 1999; Hall et al., 2011).

South-East Asian Geological Resources

Oil and Gas

Significant oil and gas accumulations occur widely in East and South-east Asia, generally in Cenozoic sedimentary basins (Fraser et al, 1997; Hutchison, 1996) (**Figure 20**), and these have contributed markedly to the economies of South-east Asian countries. The oil and gas accumulations are commonly associated with rocks of Middle and Upper Miocene age, with locally significant

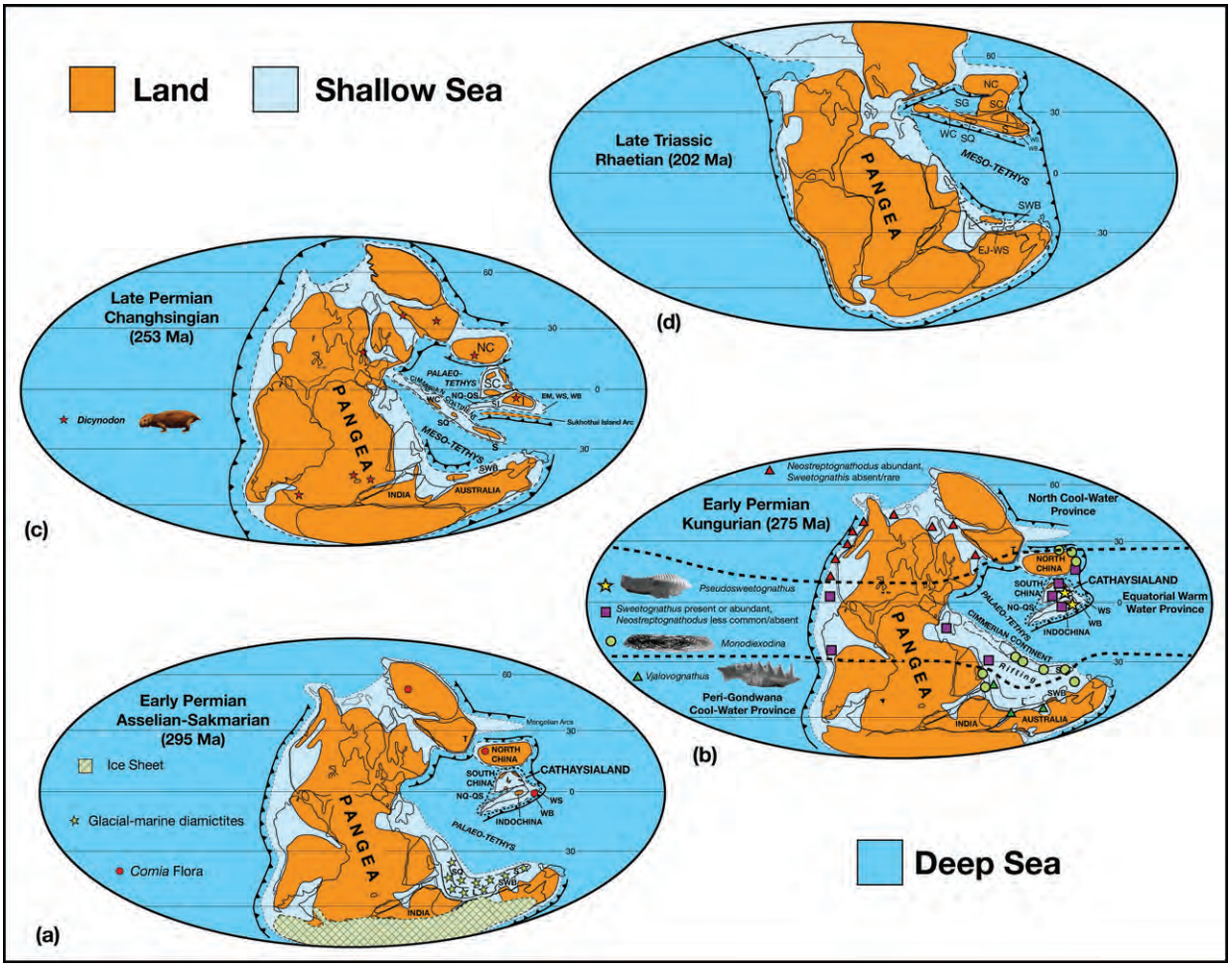


Figure 11 Palaeogeographical reconstructions of the Tethyan region for (a) the early Early Permian, (b) the late Early Permian, (c) the Late Permian, and (d) the Late Triassic, showing the relative positions of the East and South-east Asian terranes and the distribution of land and sea. Also shown are the distributions of biogeographically important biota. SC, South China; T, Tarim; I, Indochina; EM, East Malaya; WS, West Sumatra; NC, North China; SI, Simao; S, Sibumasu; WB, West Burma; QI, Qiangtang; L, Lhasa; WC, Western Cimmerian Continent; KAZ, Kazakhstan; QS, Qamdo-Simao; SG, Songpan Ganzi.

Oligocene and Pliocene occurrences. Some older Mesozoic accumulations do occur (e.g. in North Thailand), but, in general, the petroleum industry of the region is almost exclusively concerned with Cenozoic sedimentary basins, and the pre-Cenozoic is regarded as economic basement. However, in some cases, oil has migrated laterally and accumulated in fractured granitoids and other basement rocks, including Triassic limestones in the Gulf of Thailand.

Oil- and gas-bearing basins

Space constraints preclude a full detailed description of the basins and hydrocarbon fields. A number of attempts have been made to classify South-east Asian hydrocarbon-bearing basins – particularly the Cenozoic basins of the region – genetically, but these attempts have failed owing to the complex and changing pattern of compressional and extensional tectonics in South-east Asia. Basins that were previously regarded as 'typical' back-arc basins are now being interpreted by some workers as the result of major strike-slip faults. Most basins were initiated in the Eocene or Oligocene, following a major Eocene break in sedimentation. Interpretation of the genesis of the South-east Asian basins is somewhat model-dependent, and there are competing models. Whichever model one favors, there has certainly been major strike-slip faulting in the region during the Cenozoic. This strike-slip faulting can clearly be related to basin formation, for example in North and Central Thailand and the Gulf of Thailand. However, the sense and displacement of many (if not most) of these strike-slip faults are poorly known. Other basins in the region are clearly related to plate convergence and subduction; for example, the North, Central, and South Sumatra basins and the Barito Basin of Borneo appear to be back-arc basins behind the arc developed on Sundaland as the India–Australian Plate was subducted northwards. The Palawan and Sabah basins could be classified as fore-arc basins. Other basins in the region, not related genetically to strike-slip faulting or to subduction processes, have been classified variously as continental failed rifts (aulacogens), cratonic basins, or basins that have formed on or between continental fragments.

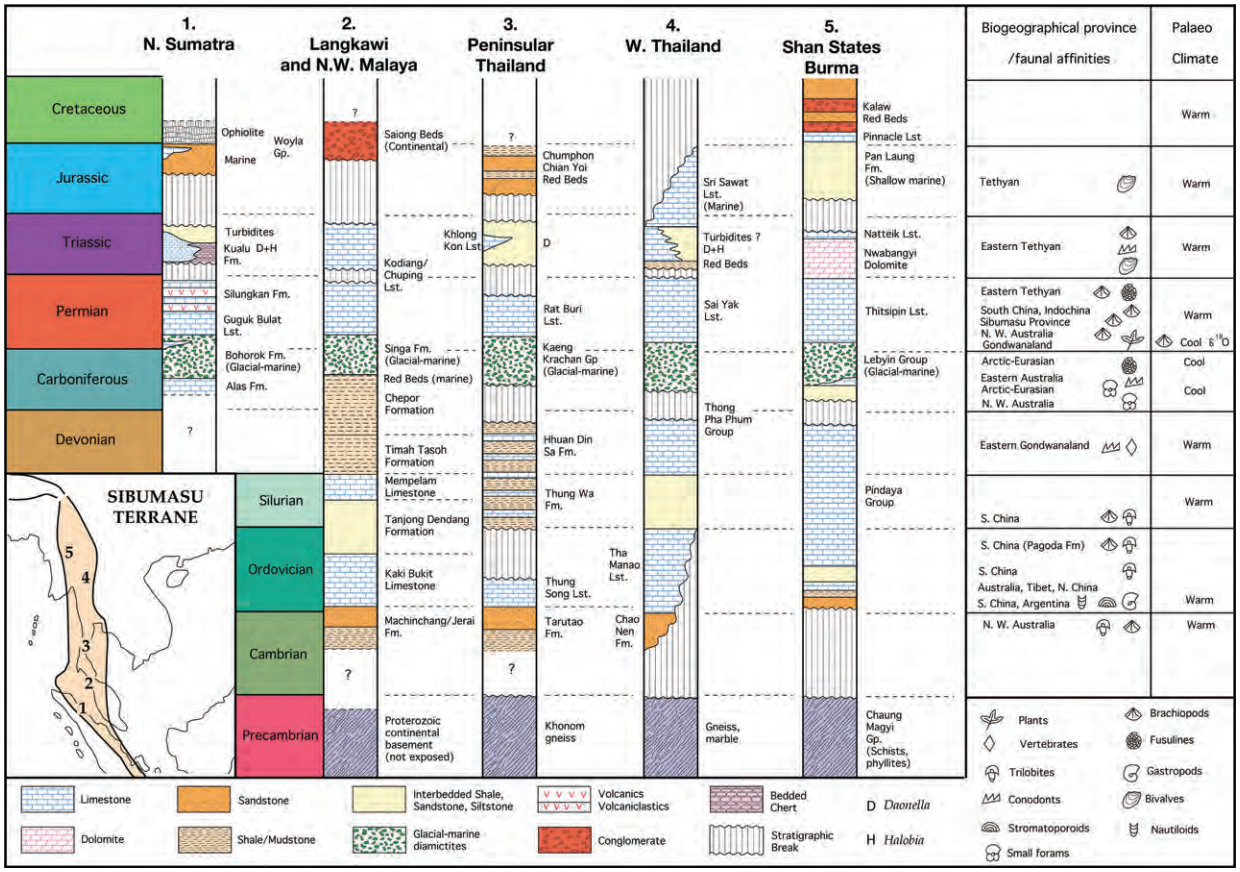


Figure 12 Representative stratigraphic columns for the Sibumasu Terrane, and faunal provinces and affinities of faunas and floras.

Minerals

The distribution of the principal mineral deposits of South-east Asia is shown in **Figure 21**. Mineralization in South-east Asia is primarily associated with the ophiolites, volcanic arcs, and granitoid plutons of the region (Hutchison, 1996).

Mineralization associated with ophiolites

Chromites are found in economic concentrations in ophiolites derived from the marginal-basin lithosphere of the Celebes, Sulu, and South China seas. Nickel sulphides and platinum also occur in dunite and serpentinite, and remobilized nickel also occurs as sulphides in veins in andesitic rocks. Deep tropical weathering of ophiolite ultramafics has resulted in nickel-bearing laterites that can be valuable ores (with a nickel oxide content of around 2.7%). Other important mineral deposits associated with ophiolites include manganese and Cyprus-type copper on pillow lavas and Besshi-type copper–iron massive sulphide deposits.

Mineralization associated with volcanic arcs

Volcanic arcs are characterized by a wide range of mineralization types, ranging from epithermal vein deposits associated with near-surface fracture systems to higher-temperature vein and dissemination deposits associated with epizonal plutons. Ore deposits include porphyry copper, Kuroko-type copper–lead–zinc, and gold and gold–silver epithermal deposits.

Non-volcanic epithermal deposits

The non-volcanic epithermal deposits of the region comprise stibnite, stibnite–gold, and stibnite–gold–scheelite mineralization, which is confined to 'Sundaland', i.e. the continental cratonic core of South-east Asia. The gold association is quite different from that of the Cenozoic volcanic arcs of the region and is confined to areas characterized by high-level felsic-to-intermediate plutons. There is commonly an important mercury association. This mineralization is younger than, and unrelated to, the tin-bearing granites of the region. Important deposits occur in western Borneo, Palawan, Sulawesi, the Malay Peninsula, Thailand, and Myanmar.

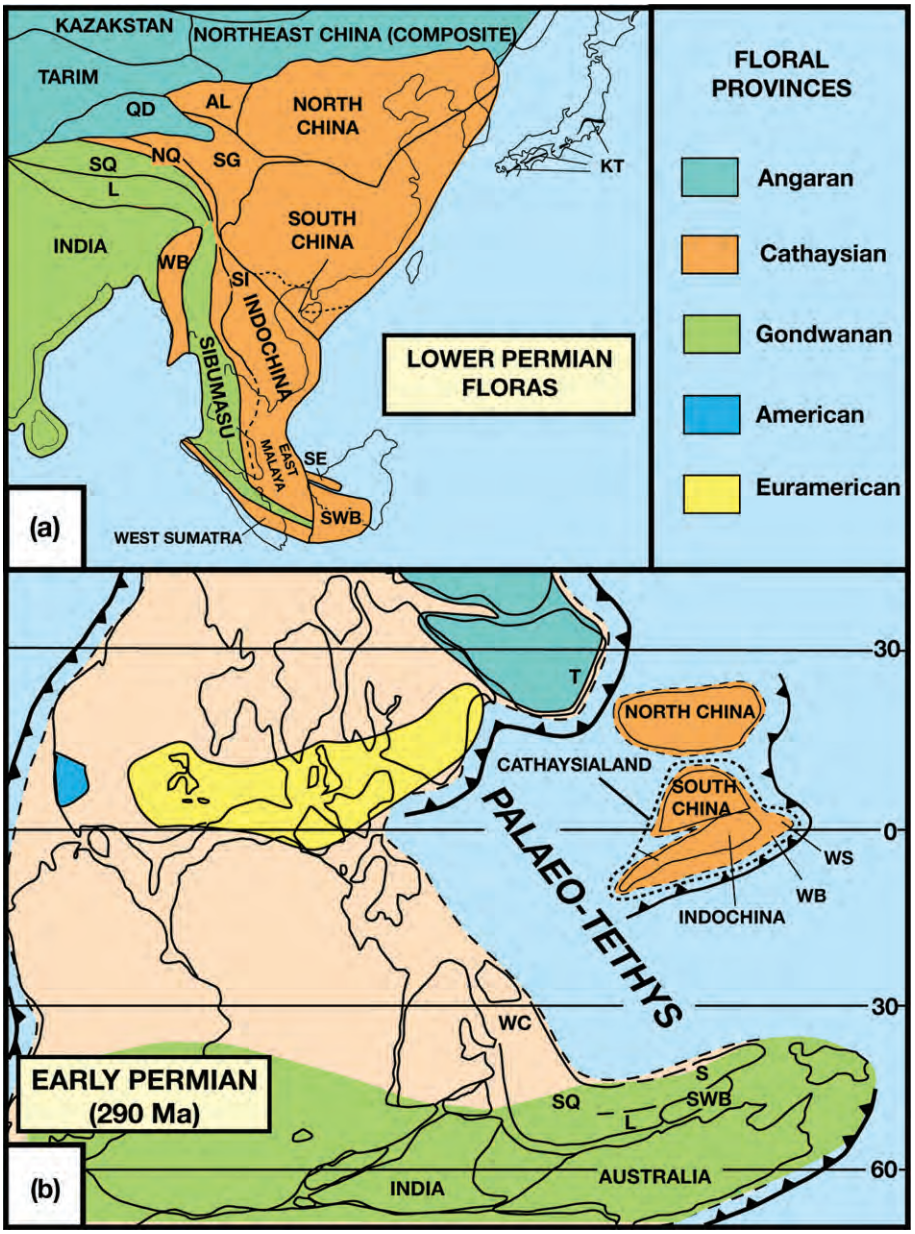


Figure 13 The distribution of Early Permian floral provinces plotted on (a) a present-day geographical map, and (b) an Early Permian palaeogeographical map. QD, Qaidam; AL, Ala Shan; SQ, South Qiantang; NQ, North Qiangtang; L, Lhasa; SG, Songpan Ganzi; WB, West Burma; SI, Simao; SE, Semitau; SWB, South West Borneo; T, Tarim; WS, West Sumatra; WC, Western Cimmerian Continent; S, Sibumasu.

Tungsten deposits

South-east Asia is a major tin–tungsten metallogenic province. The tungsten mineralization in South-east Asia is spatially and genetically related to the three main granite provinces of the region (**Figure 14**), and four types of deposit occur: hydrothermal quartz veins, scheelite skarn deposits, wolframite–scheelite–sulphide veins, and placer deposits (eluvial or marginally alluvial).

Tin deposits

During the twentieth century, 70% of the tin mined in the world came from the South-east Asian tin belt. The tin mainly occurs as cassiterite ore in Late Cenozoic alluvial placer deposits that form part of the 400 km wide South-east Asian tin belt, which extends 2800 km from Burma and Thailand through Peninsular Malaysia to the Indonesian tin islands of Bangka and Belitung (**Figure 21**). Historically, these deposits have been extremely important to the economic development of Malaysia and South-east Asia, and, altogether, 9.6 million tonnes of tin (54% of the world's total tin production) has come from this tin belt. The tin deposits of the region are associated with the major granite/granitoid bodies of South-east Asia (Schwartz et al., 1995) and formed during the emplacement of these intrusive igneous bodies, which were produced by major plate-tectonic processes as the continental blocks of

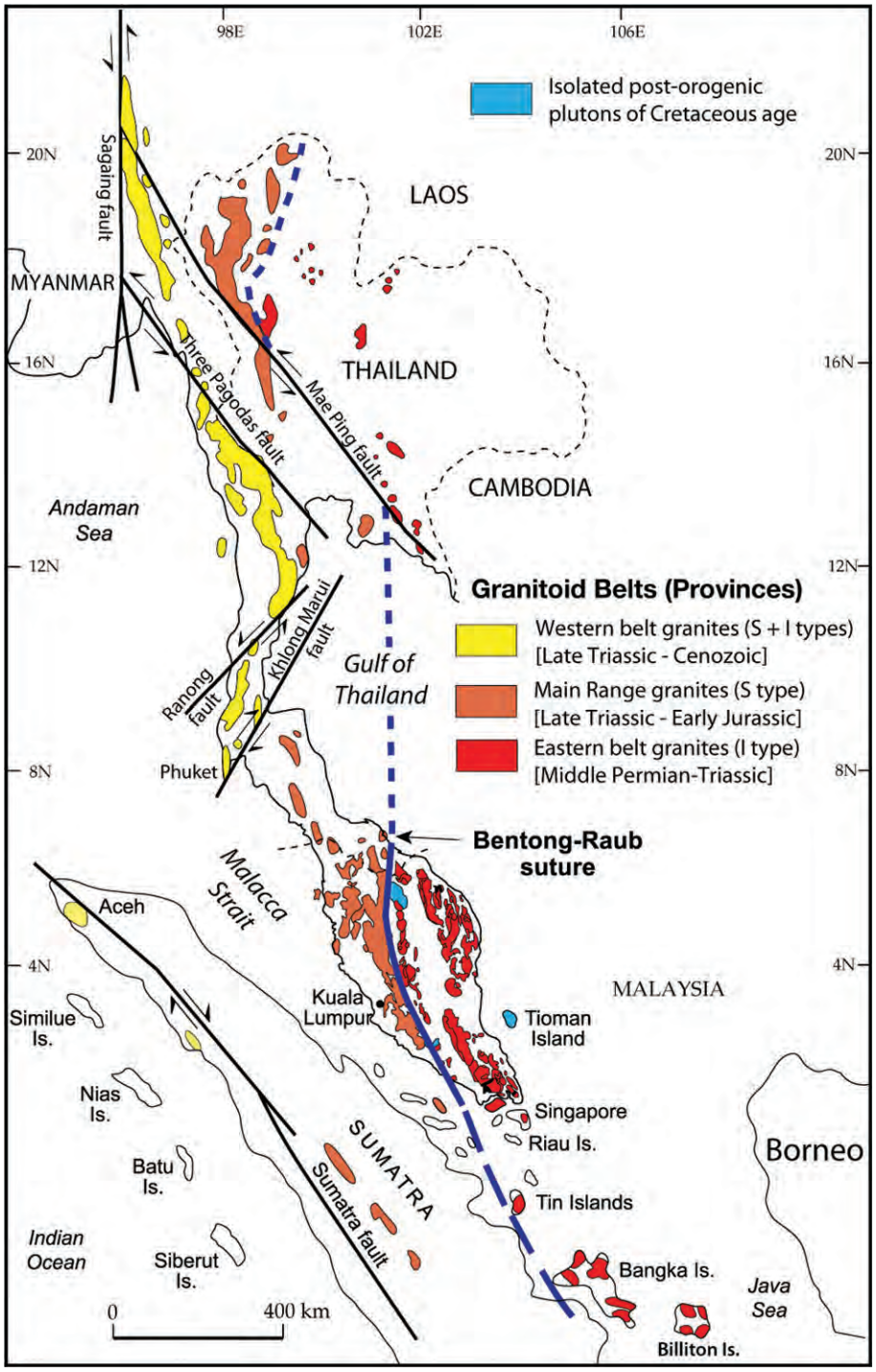


Figure 14 The three granitoid belts (provinces) of South-east Asia.

the region converged and collided with each other (Metcalfe, 2013; Searle et al., 2012). The South-east Asian granites form three major north–south belts or provinces (**Figure 14**): a Western Belt granitoid province of Late Triassic–Cenozoic (214–22 Ma) age, a central Main Range granitoid province of mainly Late Triassic and Early Jurassic (230–184 Ma) age, and an Eastern Belt granitoid province of Middle Permian-Middle Triassic (267–244 Ma) age (**Figure 14**). The Eastern province was formed by subduction on the western margins of the Indochina terrane during convergence with the Sibumasu terrane. When these two continental terranes collided, the Main Range province granites were generated (Metcalfe, 2013). The younger Western Province granites were principally formed by Tethyan subduction and terrane collisional processes. The Main Range and Eastern Belt granites of Peninsular Malaysia form parts of the central and eastern granitoid provinces, respectively. The tin mineralization occurred within

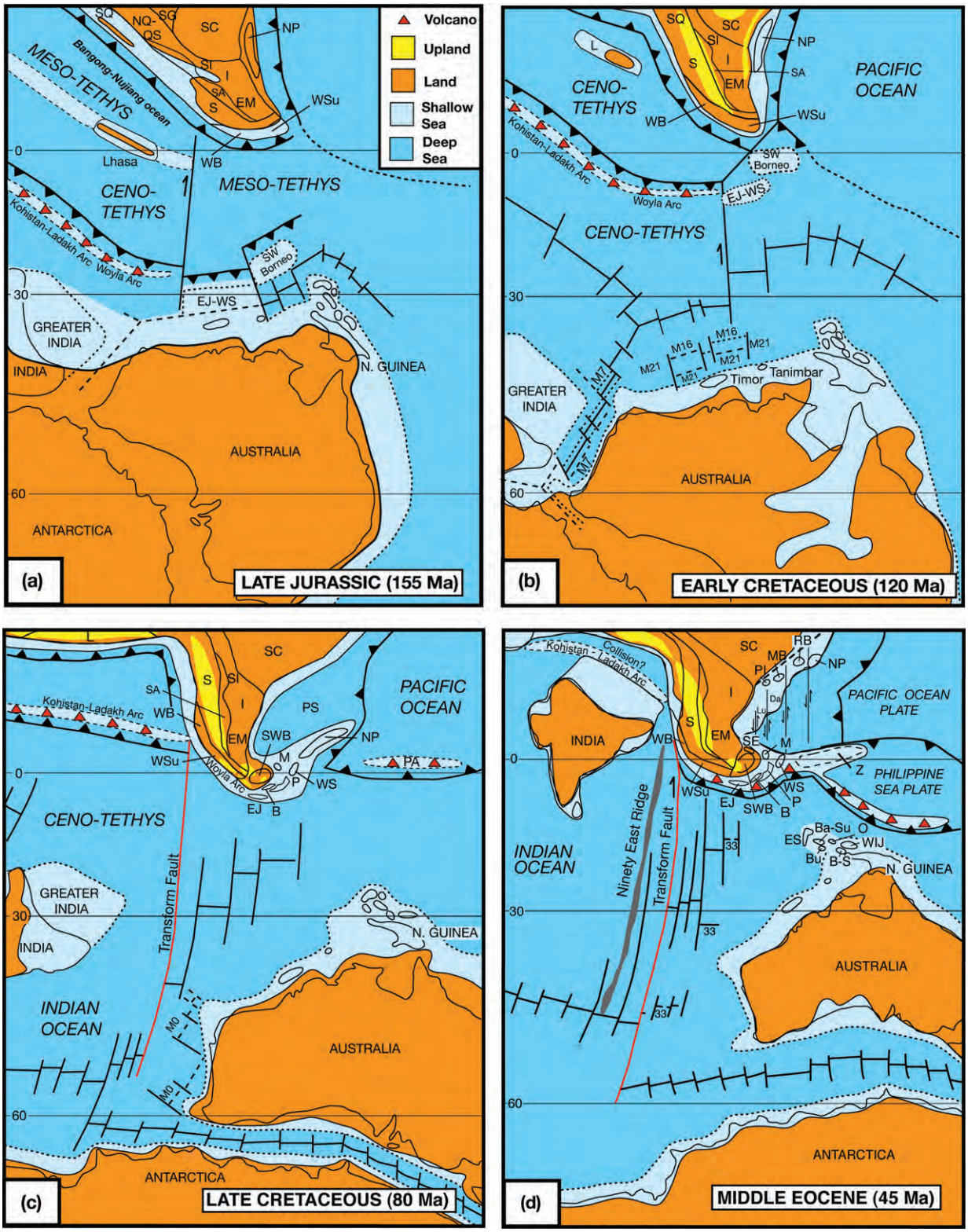


Figure 15 Palaeogeographical reconstructions for Eastern Tethys in (a) the Late Jurassic, (b) the Early Cretaceous, (c) the Late Cretaceous, and (d) the Middle Eocene, showing the distribution of continental blocks and fragments of South-east Asia–Australasia and the areas of land and sea. SG, Songpan Ganzi accretionary complex; SC, South China; QS, Qamdo-Simao; SI, Simao; QI, Qiangtang; S, Sibumasu; I, Indochina; EM, East Malaya; WSu, West Sumatra; L, Lhasa; WB, West Burma; SWB, South West Borneo; SE, Semitau; NP, North Palawan and other small continental fragments now forming part of the Philippines basement; Si, Sikuleh; M, Mangkalihat; WS, West Sulawesi; PB, Philippine Basement; PA, Incipient East Philippine Arc; PS, Proto-South China Sea; Z, Zambales Ophiolite; RB, Reed Bank; MB, Macclesfield Bank; PI, Paracel Islands; Da, Dangerous Ground; Lu, Luconia; Sm, Sumba. M numbers represent Indian Ocean magnetic anomalies.

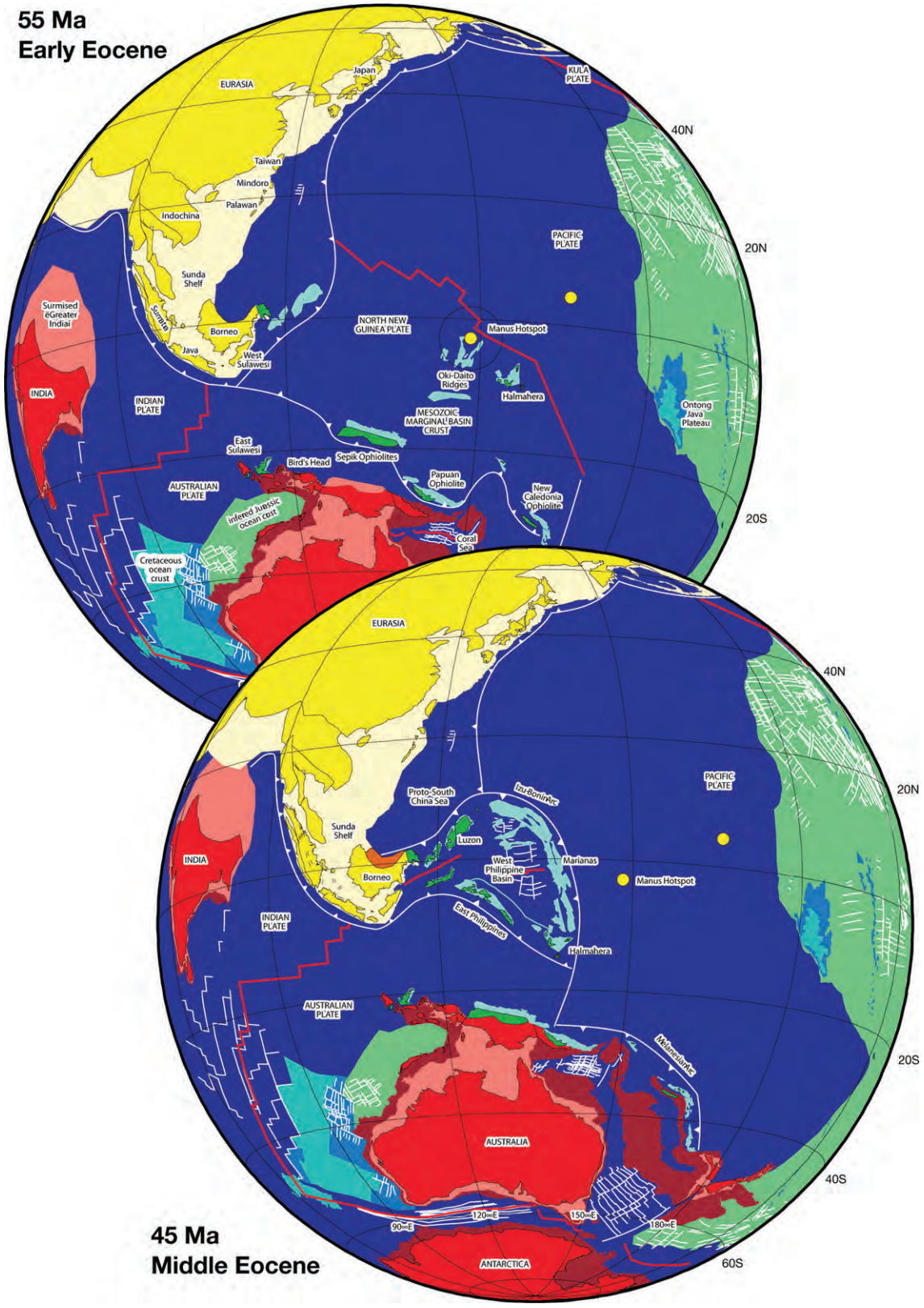


Figure 16 Plate reconstructions for South-east Asia–Australasia at 55 Ma (Early Eocene) and 45 Ma (Middle Eocene). The reconstruction at 55 Ma precedes the collision of India with Eurasia. The reconstruction at 45 Ma coincides with a major period of plate reorganization. Reproduced from Hall, R. (2002). Cenozoic geological and plate tectonic evolution of SE Asia and the SW Pacific: Computer-based reconstructions and animations. *Journal of Asian Earth Sciences* **20**, 353–434.

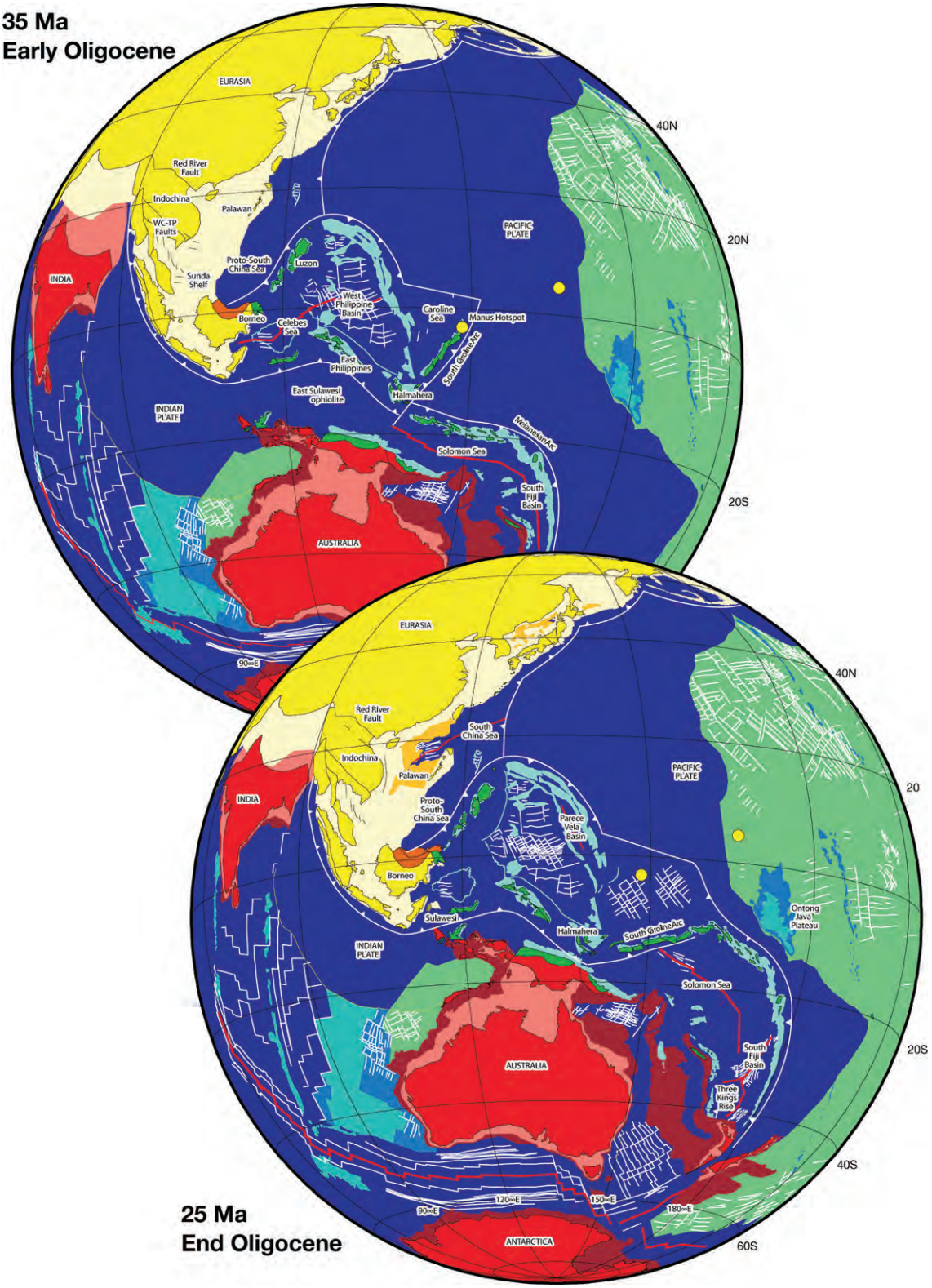


Figure 17 Plate reconstructions for South-east Asia–Australasia at 35 Ma (Early Oligocene) and 25 Ma (End Oligocene). Reproduced from Hall, R. (2002). Cenozoic geological and plate tectonic evolution of SE Asia and the SW Pacific: Computer-based reconstructions and animations. *Journal of Asian Earth Sciences* **20**, 353–434.

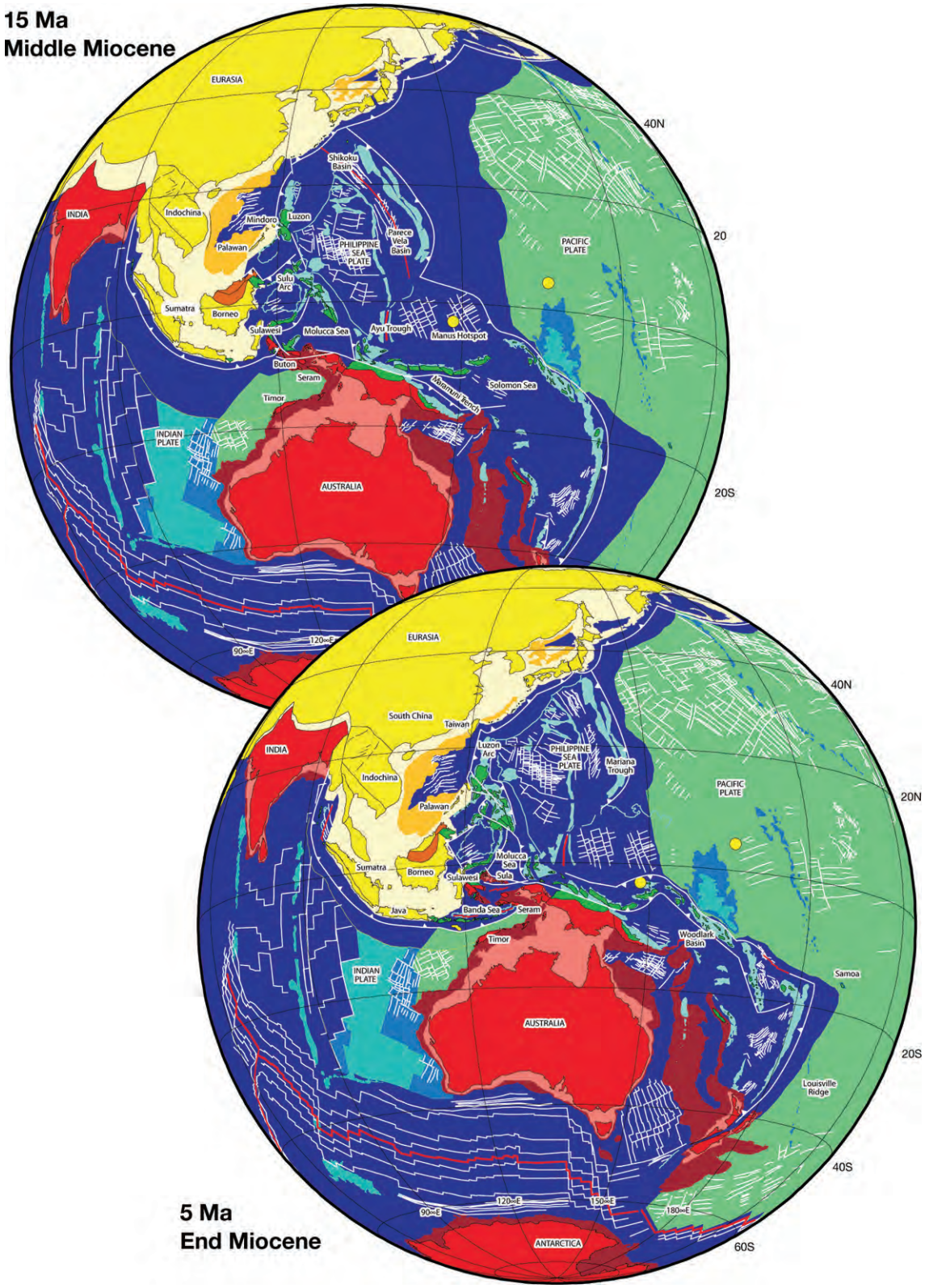


Figure 18 Plate reconstructions for South-east Asia–Australasia at 15 Ma (Middle Miocene) and 5 Ma (End Miocene). Reproduced from Hall, R. (2002). Cenozoic geological and plate tectonic evolution of SE Asia and the SW Pacific: Computer-based reconstructions and animations. *Journal of Asian Earth Sciences* **20**, 353–434.

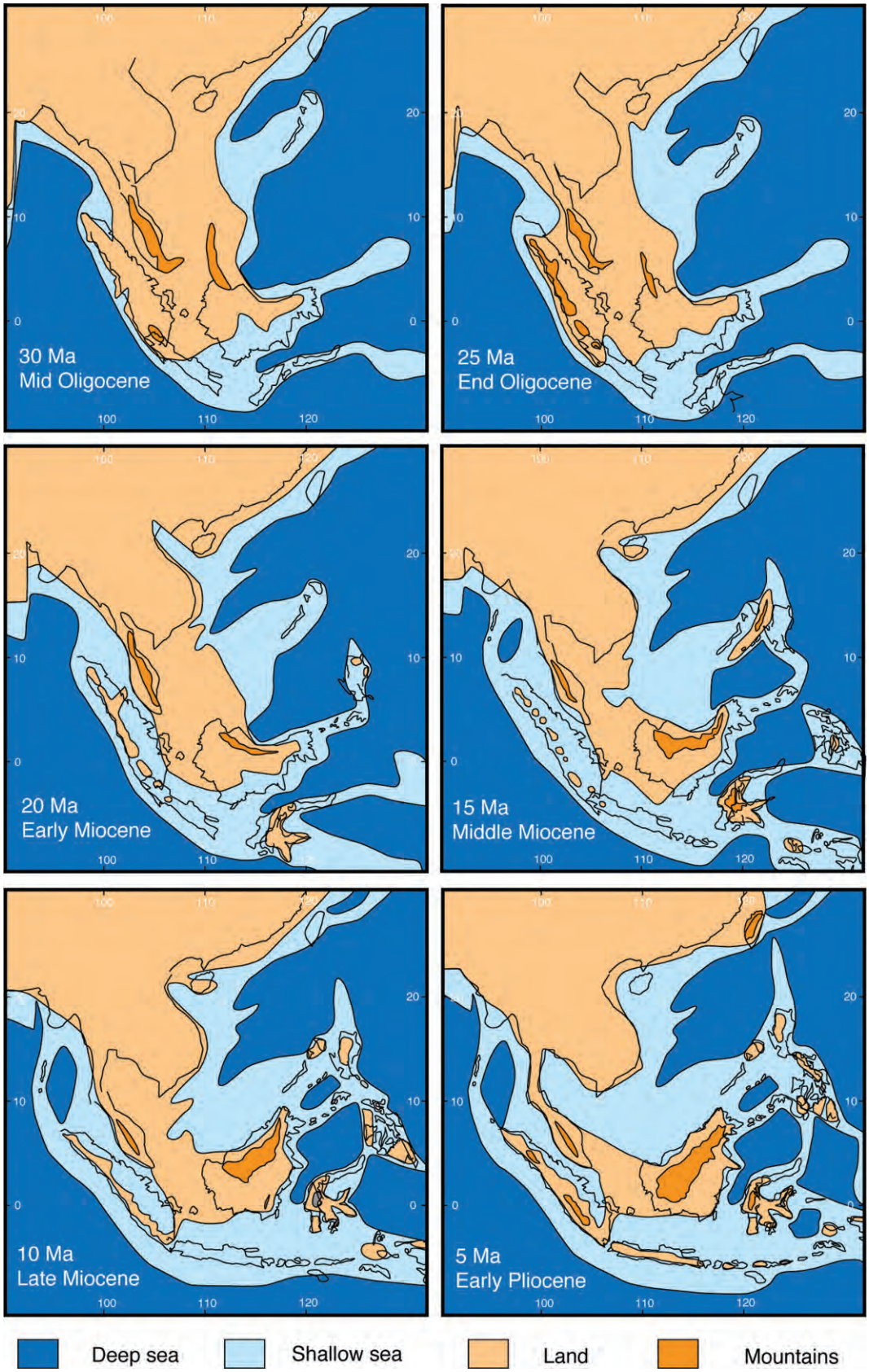


Figure 19 Reconstructions of East and South-east Asia for the last 30 Ma, showing the variation in the distributions of deep sea, shallow sea, land, and mountains.

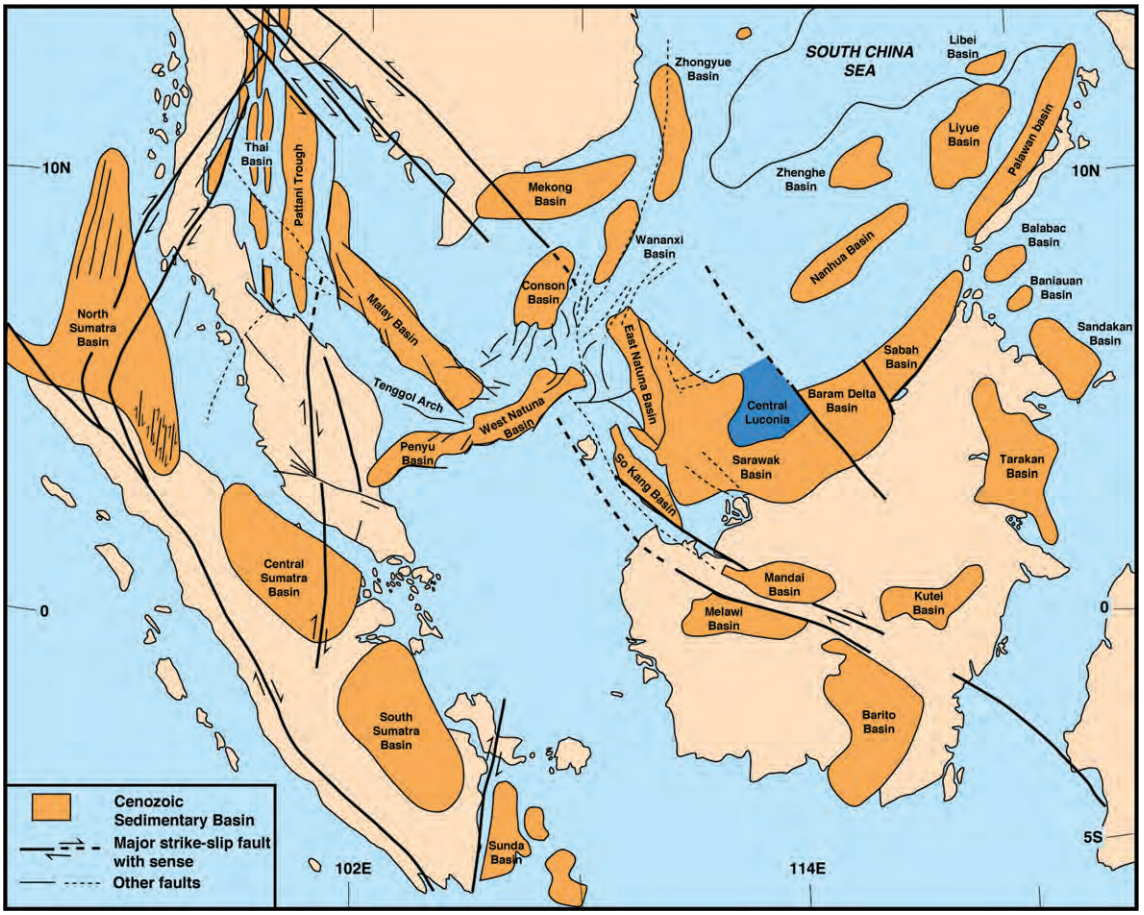


Figure 20 Oil- and gas-bearing Cenozoic sedimentary basins and major fault zones of South-east Asia.

the granites themselves and more especially in the country rocks that immediately surrounded them. Subsequent to mineralization, the granites and the country rocks into which they were intruded were uplifted and eroded, and the tin ore was deposited and concentrated in river alluvium.

In addition to placer tin deposits there are also a wide variety of primary tin deposits. However, only two major primary tin fields have led to significant deep-mine production: the Sungei Lembing Mine near Kuantan, Pahang, Peninsular Malaysia, and Kelapa Kampit on Billiton Island. The lodes of these mines are developed in contact-metamorphosed Carboniferous–Permian argillaceous sedimentary and andesitic tuffaceous rocks within aureoles around high-level Permian–Triassic granitoids.

Mississippi valley-type epigenetic deposits

Stratiform deposits of predominantly galena and sphalerite are most commonly hosted in shallow-marine carbonate-dominated sequences on the rims of large sedimentary basins where changes in salinity caused deposition. The continental regions of South-east Asia, with their extensive Palaeozoic carbonate platforms, are suited to Mississippi Valley-type epigenetic deposits, and some have been identified. One of the best-known deposits in the region, now considered to be of this type, is at the Bawdwin lead–zinc mine north-north-east of Rangoon, Myanmar. Mining here can be traced back to 1412.

Iron ore deposits

Iron ore deposits are widespread in the region, but significant production has been restricted to three areas: South-east China (Hainan, Kwangtung, Fukien), the Malay Peninsula, and the Philippines. The deposits are predominantly of contact pyrometasomatic type and are associated with plutons emplaced at a wide range of depths. Peninsular Malaysia was a major producer of iron ore, with production reaching 7 million tonnes in 1964. However, all major deposits have now been worked out.

Coal and lignite deposits

South-east Asian coal deposits are almost entirely confined to deposits of sub-bituminous and lignitic coal in relatively small Cenozoic terrestrial basins. There are minor occurrences of bituminous coal and anthracite in Palaeozoic and Mesozoic formations of the older Sundaland continental core of the region, but these are generally uneconomic except for deposits of Triassic anthracite mined in northern Vietnam.

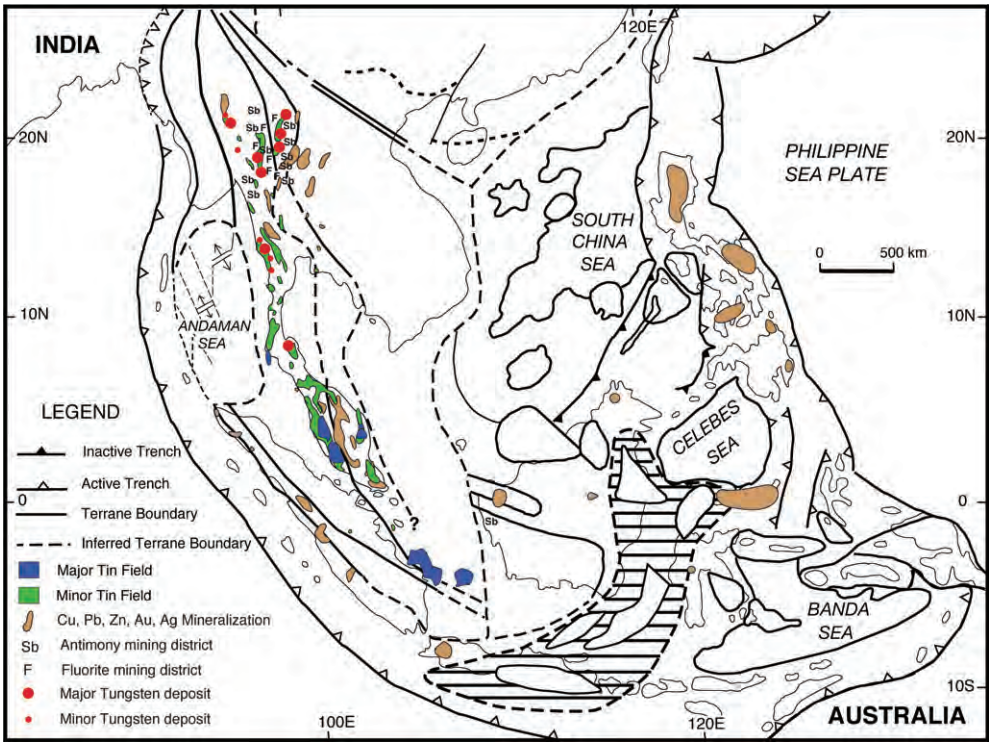


Figure 21 Distribution of the principal mineral deposits of South-east Asia.

Diamonds, sapphires, and other gems

Alluvial diamonds with no obvious sources ('headless placers') are found in several areas of South-east Asia, including Myanmar, southern Thailand (Phuket), Sumatra, and Kalimantan (**Figure 8**). These deposits occur in relatively young geological terranes, in contrast to the Archaean or Proterozoic terranes that host most primary diamond deposits and their associated alluvial workings.

Significant quantities of diamond have been recovered from two areas in Myanmar: Momeik in the northern part of the country and Theindaw in the southern part. The Momeik diamonds are recovered during mining of gemstone gravels; the Theindaw and Phuket diamonds are by-products of tin dredging. These sites, and those in Sumatra, lie within the Sibumasu terrane, which was detached from north-western Australia in Palaeozoic times and drifted northwards to become part of South-east Asia. The Myanmar–Thailand diamonds may be derived either from lamproitic intrusives within this terrane or secondarily from Permian glacial–marine sediments. The Kalimantan diamonds occurring in SW Borneo placer deposits without any likely local source, have geochemical and isotope signatures similar to Australian diamonds (Smith et al., 2009; Taylor et al. 1990).

Sapphire, ruby, zircon, and spinel gemstones and jade are worked at a number of localities in South-east Asia from alluvial and eluvial gravels derived from nearby outcrops of extensive Pliocene–Pleistocene high-alkali basalt fields. The basalts are usually basinite and nephelinite. The main centres for the mining of these gemstones are at Pailin, Bokeo, and Rovieng in Cambodia, Chanthaburi, Trat, and Kancharaburi in Thailand, south of Haikou on Hainan Island, in the Mogok and Lonkin areas of Myanmar, and in Taiwan. Most of the South-east Asian rubies come from the Trat area of Thailand, which supplies 70% of the world's production. Ruby, sapphire, and spinel are mined in the Mogok area of Myanmar from placers and eluvial deposits derived from marbles and calc-silicate migmatites associated with alaskite (Mogok Gneiss). Jadeite is mined in the Lonkin area of Myanmar, which is the world's foremost source of high-grade jadeite.

References

Ali JR and Aitchison JC (2007) When and where did India and Asia collide? *Journal of Geophysical Research* 112(B05423): 1–19.

Ali JR and Aitchison JC (2008) Gondwana to Asia: Plate tectonics, paleogeography and the biological connectivity of the Indian sub-continent from the Middle Jurassic through latest Eocene (166–35 Ma). *Earth Science Reviews* 88: 145–166.

Ampaiwan T, Hisada K, and Charusiri P (2009) Lower Permian glacially influenced deposits in Phuket and adjacent islands, peninsular Thailand. *Island Arc* 18: 52–68.

Battail B (2009) Late Permian dicynodont fauna from Laos. In: Buffetaut E, Cuny G, Le Loeuff J, and Suteethorn V (eds.) *Late Palaeozoic and Mesozoic ecosystems in SE Asia*315: pp. 33–40. London: The Geological Society Special Publications.

Chang C, Chen N, Coward MP, Deng W, Dewey JF, Gansser A, Harris NBW, Jin C, Kidd WSF, Leeder MR, Li H, Lin J, Liu C, Mei H, Molnar P, Pan Y, Pan Y, Pearce JA, Shackleton RM, Smith AB, Sun Y, Ward M, Watts DR, Xu J, Xu R, Yin J, and Zhang Y (1986) Preliminary conclusions of the Royal Society and Academia Sinica 1985 geotraverse of Tibet. *Nature* 323: 501–507.

Colwell JB, Stagg HMJ, Symonds PA, Willcox JB, and O'Brien GW (compilers for AGSO North West Shelf Study Group) (1994) Deep reflections on the North West Shelf: changing perceptions of basin formation. In: Purcell PG and Purcell RR (eds.) *The Sedimentary Basins of Western Australia*, pp. 63–76. Petroleum Exploration Society of Australia.
Dewey JF and Burke K (1973) Tibetan, Variscan and Precambrian basement reactivation: Products of continental collision. *Journal of Geology* 81: 683–692.
Dewey JF, Shackleton RM, Chang C, and Sun Y (1988) The tectonic evolution of the Tibetan plateau. *Philosophical Transactions of the Royal Society of London. Series A* 327: 379–413.
Fielding CR, Frank TD, Birgenheier LP, Rygel MC, Jones AT, and Roberts J (2008) Stratigraphic imprint of the Late Palaeozoic Ice Age in eastern Australia: A record of alternating glacial and nonglacial climate regime. *Journal of the Geological Society of London* 165: 129–140.
Fraser AJ, Matthews SJ, and Murphy RW (eds.) (1997) *Petroleum geology of southeast Asia*. London: Geological Society of London Special Publication 126.
Hall R (2002) Cenozoic geological and plate tectonic evolution of SE Asia and the SW Pacific: Computer-based reconstructions, model and animations. *Journal of Asian Earth Sciences* 20: 353–431.
Hall R (2009) The Eurasia SE Asian Margin as a Modern Example of an Accretionary Orogen. In: Cawood PA and Kroner A (eds.) *Earth Accretionary Systems in Space and Time*, pp. 351–372. The Geological Society London, Special Publication No. 318.
Hall R (2012) Late Jurassic–Cenozoic reconstructions of the Indonesian region and the Indian ocean. *Tectonophysics* 570–571: 1–41.
Hall R, Clements B, and Smyth HR (2009) Sundaland: Basement character, structure and plate tectonic development. *Proceedings, Indonesian Petroleum Association Thirty-Third Annual Convention and Exhibition*, May 2009, IPA09-G-134, pp. 1–27.
Hall R, Cottam MA, and Wilson MEJ (eds.) (2011) *The SE Asian gateway: History and tectonics of the Australia-Asia collision*. London: Geological Society of London Special Publication 355.
Hall R and Holloway JD (eds.) (1998) *Biogeography and geological evolution of South-east Asia*. Amsterdam: Backhuys Publishers.
Hall R and Sevastjanova I (2012) Australian crust in Indonesia. *Australian Journal of Earth Sciences* 59: 827–844.
Hall R, van Hattum MCA, and Spakman W (2008) Impact of India–Asia collision on SE Asia: the record in Borneo. *Tectonophysics* 451: 366–389.
Hutchison CS (1989) *Geological evolution of South-east Asia*. Oxford: Clarendon Press.
Hutchison CS (1996) *South-east Asian oil, gas, coal and mineral deposits*. Oxford: Clarendon Press.
Klootwijk C (1996) Phanerozoic configurations of greater Australia: Evolution of the north west shelf: Part two. Palaeomagnetic and geologic constraints on reconstructions. *AGSO Record* 1996(52): 1–85.
Li P, Rui G, Junwen C, and Ye G (2004) Paleomagnetic analysis of eastern Tibet: Implications for the collisional and amalgamation history of the three rivers region, SW China. *Journal of Asian Earth Sciences* 24: 291–310.
Metcalfe I (1988) Origin and assembly of Southeast Asian continental terranes. In: Audley-Charles MG and Hallam A (eds.) *Gondwana and Tethys*, pp. 101–118. London: The Geological Society Special Publication No. 37.
Gondwana dispersion and Asian accretion. Metcalfe I (ed.) (1994) In: *Special Issue of Journal of Southeast Asian Earth Sciences*9: 303–461.
Metcalfe I (1998) Palaeozoic and Mesozoic geological evolution of the SE Asian region: Multidisciplinary constraints and implications for biogeography. In: Hall R and Holloway JD (eds.) *Biogeography and geological evolution of SE Asia*, pp. 25–41. Amsterdam, The Netherlands: Backhuys Publishers.
Metcalfe I (ed.) (1999) *Gondwana dispersion and Asian accretion: IGCP 321 final results volume*. Rotterdam: A. A. Balkema.
Metcalfe I (2011) Tectonic framework and Phanerozoic evolution of Sundaland. *Gondwana Research* 19: 3–21.
Metcalfe I (2013) Gondwana dispersion and Asian accretion: Tectonic and palaeogeographic evolution of eastern Tethys. *Journal of Asian Earth Sciences* 66: 1–33.
Suture zones of East and Southeast Asia. Metcalfe I and Allen MB (eds.) (2000) *Special Issue of Journal of Asian Earth Sciences* 18: 635–808.
Metcalfe I, Smith JMB, Morwood M, and Davidson I (eds.) (2001) *Faunal and floral migrations and evolution in SE Asia–Australasia*. Lisse: A A Balkema.
Powell CM and Conaghan PG (1973) Plate tectonics and the Himalayas. *Earth and Planetary Science Letters* 20: 1–12.
Rong J-Y, Boucot AJ, Yang-Zheng Su, and Strusz DL (1995) Biogeographical analysis of late Silurian brachiopod faunas, chiefly from Asia and Australia. *Lethaia* 28: 39–60.
Searle MP, Whitehouse MJ, Robb LJ, Ghani AA, Hutchison CS, Sone M, Ng W-P, Roselee MH, Chung S-L, and Oliver GJH (2012) Tectonic evolution of the Sibumasu-Indochina terrane collision zone in Thailand and Malaysia: Constraints from new U -Pb zircon chronology of SE Asian tin granitoids. *Journal of the Geological Society* 169: 489–500.
Sevastjanova I, Clements B, Hall R, Belousova EA, Griffin WL, and Pearson N (2011) Granitic magmatism, basement ages, and provenance indicators in the Malay peninsula: Insights from detrital zircon U–Pb and Hf-isotope data. *Gondwana Research* 19: 1024–1039.
Shen S-Z, Zhang H, Shi GR, Li W-Z, Xie J-F, Mu L, and Fan J-X (2013) Early Permian (Cisuralian) global brachiopod palaeobiogeography. *Gondwana Research* http://dx.doi.org/10.1016/j.gr.2012.05.017.
Shi GR and Archbold NW (1998) Permian marine biogeography of SE Asia. In: Hall R and Holloway JD (eds.) *Biogeography and geological evolution of SE Asia*, pp. 57–72. Amsterdam, The Netherlands: Backhuys Publishers.
Smith CB, Bulanova GP, Kohn SC, Milledge HJ, Hall AE, Griffin BJ, and Pearson DG (2009) Nature and genesis of Kalimantan diamonds. *Lithos* 112: 822–832.
Sone M and Metcalfe I (2008) Parallel Tethyan sutures in mainland SE Asia: new insights for Palaeo-Tethys closure. *Compte Rendus Geoscience* 340: 166–179.
Stauffer PH and Lee CP (1989) Late Palaeozoic glacial marine facies in southeast Asia and its implications. *Geological Society of Malaysia Bulletin* 20: 363–397.
Stauffer PH and Mantajit J (1981) Late Palaeozoic tilloids of Malaya, Thailand and Burma. In: Hambrey MJ and Harland WH (eds.) *Earth's pre-Pleistocene glacial record*, pp. 331–337. Cambridge: Cambridge University Press.
Schwartz MO, Rajah SS, Askury AK, Putthapiban P, and Djaswadi S (1995) The southeast Asian Tin belt. *Earth-Science Reviews* 38: 95–293.
Tapponnier P, Peltzer G, Le Dain AY, Armijo R, and Cobbold P (1982) Propagating extrusion tectonics in Asia: new insights from simple experiments with plasticine. *Geology* 10: 611–616.
Tapponnier PG, Peltzer G, and Armijo R (1986) On the mechanisms of the collision between India and Asia. In: Coward MP and Ries AC (eds.) *Collision Tectonics*, pp. 115–157. The Geological Society London, Special Publication No. 19.
Taylor WR, Jaques AL, and Ridd M (1990) Nitrogen-defect aggregation characteristics of some Australasian diamonds: Time-temperature constraints on the source regions of pipe and alluvial diamonds. *American Mineralogist* 75: 1290–1310.
Ueno K (2003) The Permian fusulinoidean faunas of the Sibumasu and Baoshan blocks: Their implications for the paleogeographic and paleoclimatologic reconstruction of the Cimmerian continent. *Palaeogeography, Palaeoclimatology, Palaeoecology* 193: 1–24.
Van Oosterzee P (1997) *Where worlds collide: The Wallace line*. Kew, Victoria: Reed.
Wang Z and Tan X (1994) Palaeozoic structural evolution of Yunnan. *Journal of Southeast Asian Earth Sciences* 9: 345–348.
Yin A (2010) Cenozoic tectonic evolution of Asia: a preliminary synthesis. *Tectonophysics* 488: 293–325.
Zhao W, Nelson KD, Che J, Borwn LD, Xu Z, and Kuo JT (1993) Deep seismic reflection evidence for continental underthrusting beneath southern Tibet. *Nature* 366: 557–559.
Zhou H-W and Murphy MA (2005) Tomographic evidence for wholesale underthrusting of India beneath the entire Tibetan plateau. *Journal of Asian Earth Sciences* 25: 445–457.
Zhu D-C, Zhao Z-D, Niu Y, Dilek Y, and Mo X-X (2011) Lhasa terrane in southern Tibet came from Australia. *Geology* 39: 727–730.
Zhu D-C, Zhao Z-D, Niu Y, Dilik Y, Hou Z-Q, and Mo X-X (2012) The origin and pre-Cenozoic evolution of the Tibetan plateau. *Gondwana Research* 23: 1429–1454.